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PHYSICS

1) **Assertion :** For a particle performing S.H.M., its speed decreases as it goes away from the mean position.
Reason : In S.H.M., the acceleration changes with time.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

2) For a particle of mass m executing SHM with angular frequency ω, the kinetic energy is given by $k = k_0 \cos^2 \omega t$. The equation of its displacement can be

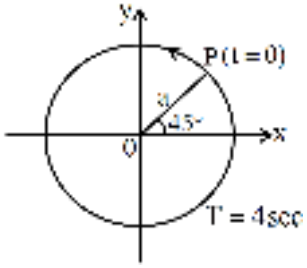
(1) $\left(\frac{k_0}{m\omega^2}\right)^{1/2} \sin(\omega t)$
(2) $\left(\frac{2k_0}{m\omega^2}\right)^{1/2} \sin(\omega t)$
(3) $\left(\frac{2\omega^2}{mk_0}\right)^{1/2} \sin(\omega t)$
(4) $\left(\frac{k_0}{2m\omega^2}\right)^{1/2} \sin(\omega t)$

3) A particle of mass m in a unidirectional potential field have potential energy $U(x) = a + 2bx^2$; where a and b are positive constants. Find time period of oscillation.

(1) $2\pi\sqrt{\frac{2b}{m}}$
(2) $2\pi\sqrt{\frac{m}{2b}}$
(3) $\pi\sqrt{\frac{m}{b}}$
(4) $\pi\sqrt{\frac{b}{m}}$

4) The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the figure -

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x-projection of the radius vector of particle P is :-

(1) $x = a\cos\left(\pi t + \frac{\pi}{4}\right)$

(2) $x = a\sin\left(\pi t + \frac{\pi}{4}\right)$

(3) $x = a\cos\left(\frac{2\pi t}{4} + \frac{\pi}{4}\right)$

(4) $x = a\sin\left(\frac{2\pi t}{4} + \frac{\pi}{4}\right)$

5) A mass M is suspended form a spring of negligible mass. The spring is pulled a little and then released so that the mass executes simple harmonic oscillations with a time period T. If the mass is increased by m then the time period becomes $\left(\frac{5}{4}T\right)$. The ratio of $\frac{m}{M}$ is :

(1) 9/16

(2) 25/16

(3) 4/5

(4) 5/4

6) The frequency of tuning forks A and B are respectively 3% more and 2% less than the frequency of tuning fork C. When A and B are simultaneously excited, 5 beats per second are produced. Then the frequency of the tuning fork 'A' (in Hz) is:

(1) 98

(2) 100

(3) 103

(4) 105

7) Two coherent sources of intensities I_1 and I_2 produce an interference pattern. The maximum intensity in the interference pattern will be –

(1) $I_1 + I_2$

(2) $I_1^2 + I_2^2$

(3) $(I_1+I_2)^2$

(4) $\left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

8) Four simple harmonic vibrations

$y_1 = 8\cos\omega t$; $y_2 = 4\cos\left(\omega t + \frac{\pi}{2}\right)$

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$y_3 = 2\cos(\omega t + \pi)$; $y_4 = \cos\left(\omega t + \frac{3\pi}{2}\right)$
and superposed on each other. The resulting amplitude and phase are respectively :

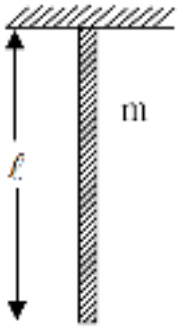
(1) $\sqrt{45}$ and $\tan^{-1}(1/2)$

(2) $\sqrt{45}$ and $\tan^{-1}(1/3)$

(3) $\sqrt{75}$ and $\tan^{-1}(1/2)$

(4) $\sqrt{75}$ and $\tan^{-1}(1/3)$

9) A uniform rope of mass m and length hangs from a ceiling as shown in figure. Find out time



taken by a transverse wave to travel the full length of rope.

(1) $t = \sqrt{\frac{\ell}{g}}$

(2) $t = \sqrt{\frac{g}{\ell}}$

(3) $t = 2\sqrt{\frac{\ell}{g}}$

(4) $t = \frac{1}{2}\sqrt{\frac{\ell}{g}}$

10) A string with a linear mass density of 4×10^{-3} kg/m is under tension of 360 N and is fixed at both ends. One of its resonance frequencies is 375 Hz. The next higher resonance frequency is 450 Hz. The mass of the string in gram is :-

(1) 16

(2) 8

(3) 24

(4) 4

11) A number of tuning forks are arranged in the order of increasing frequency and any two successive tuning forks produce 4 beats/s, when sounded together. If the last TF has a frequency octave higher than that of the first TF and frequency of the first T.F. is 256 Hz then the number of tuning forks is :-

(1) 63

(2) 64

(3) 65

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(4) 66

12) A student is performing experiment with resonance tube apparatus in Physics lab to find the speed of sound at room temperature. He got first two resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature.

(1) 348 m/s

(2) 358 m/s

(3) 332 m/s

(4) 322 m/s

13) Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C. The temperature rise of ethyl alcohol will be (specific heat of ethyl alcohol is one half of the specific heat of water) :-

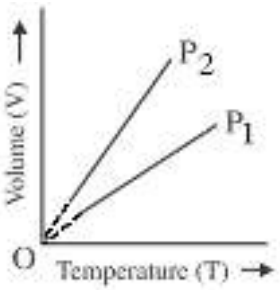
(1) 12.5°C

(2) 25°C

(3) 50°C

(4) It depends on the rate of energy transfer

14) For a perfect gas, two pressures P_1 and P_2 are shown in figure. The graph shows:



(1) $P_1 > P_2$

(2) $P_1 < P_2$

(3) $P_1 = P_2$

(4) Insufficient data to draw any conclusion

15) In an adiabatic change, then pressure and temperature of a diatomic gas are related as $P \propto T^C$ where C equal to :-

(1) $\frac{5}{3}$

(2) $\frac{2}{5}$

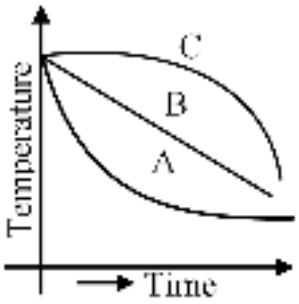
(3) $\frac{3}{5}$

(4) $\frac{7}{2}$

16) A block of steel heated to 100°C is left in a room to cool. Which of the curves shown in fig.

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represents the correct behavior ?

(1) A
(2) B
(3) C
(4) None of these

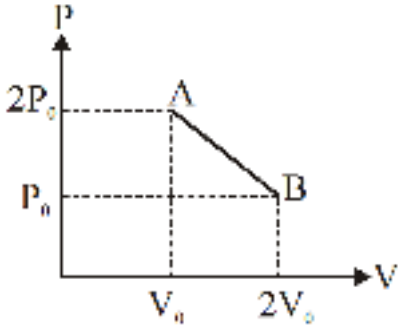
17) 8 kg of solid material is heated from 20°C to 120°C with the addition of 800 kJ of heat in a furnace. What will be it specific heat ?

(1) 1 kJ/kg °C
(2) 2 kJ/kg °C
(3) 3 kJ/kg °C
(4) 4 kJ/kg °C

18) On heating a liquid of coefficient of cubical expansion γ in a container having coefficient of linear expansion $\frac{\gamma}{3}$, the level of liquid in the container will :

(1) Rise
(2) Remains stationary
(3) Fall
(4) None of these

19) 'n' moles of an ideal gas undergoes a process A → B as shown in the figure. The maximum

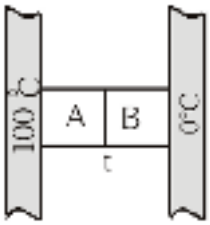


temperature of the gas during the process will be :

(1) $\frac{9P_0V_0}{nR}$
(2) $\frac{9P_0V_0}{4nR}$
(3) $\frac{3P_0V_0}{2nR}$

(4) $\frac{9P_0V_0}{2nR}$

20) Two metal cubes A and B of same size are arranged as shown in the figure. The extreme ends of the combination are maintained at the indicated temperatures. The arrangement is thermally insulated. The coefficients of thermal conductivity of A and B are 300 W/m °C and 200 W/m °C, respectively. After steady state is reached, the temperature t of the interface will be :-



(1) 45°C
(2) 90°C
(3) 30°C
(4) 60°C

21) 50 g of ice at 0°C is mixed with 200 gm of water at 0°C. 6 kcal heat is given to system [ice + water]. Find the temperature of the system.

(1) 10°C
(2) 8°C
(3) 20°C
(4) 4°C

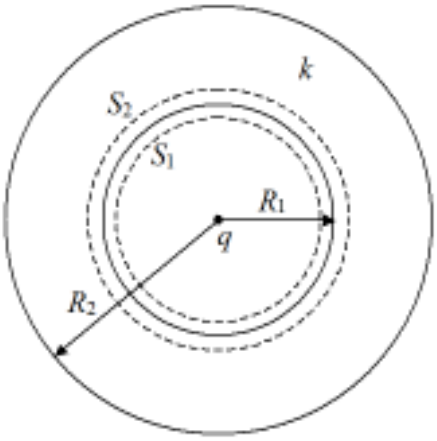
22) The rate radiation emitted by a star A is 10,000 times that of the sun. If the surface temperatures of the sun and star A are 6000 K and 2000 K respectively, the ratio of the radii of the star A and the sun is :-

(1) 300 : 1
(2) 600 : 1
(3) 900 : 1
(4) 1200 : 1

23) A positive charge q is kept at the center of a thick shell of inner radius R_1 and outer radius R_2 which is made up of conducting material. If ϕ_1 is flux through closed gaussian surface S_1 whose radius is just less than R_1 and ϕ_2 is flux through closed gaussian surface S_2 whose radius is just

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greater than R_1, then:-

(1) $\phi_1 > \phi_2$

(2) $\phi_2 > \phi_1$

(3) $\phi_1 = \phi_2 = \frac{q}{\varepsilon_0}$

(4) $\phi_1 = \phi_2 = \frac{kq}{\varepsilon_0}$

24) Two identical charged spheres are suspended from a rigid support by strings of equal lengths. The strings make an angle of 30° with each other. When system suspended in a liquid of density 0.8 g cm^{-3}, the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3}, the dielectric constant of the liquid is :-

(1) 4

(2) 3

(3) 2

(4) 1

25) Two protons release from mutual distance r, find the maximum velocity they can achieve (assume charge of proton = e & mass of proton = m_p) :-

(1) $v = e\sqrt{\frac{K}{m_p r}}$

(2) $v = e\sqrt{\frac{2K}{m_p r}}$

(3) $v = e\sqrt{\frac{K}{2m_p r}}$

(4) None of these

26) Two charges q_1 and q_2, when combined gives a total charge of 6 µC. When they are separated by 3 m, the repulsive electric force between them is 8 mN. What is the value of q_1?

(1) 8 µC

(2) 4 μC

(3) 1 μC

(4) Zero

27) Let $\rho(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point P inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is :-

(1) zero

(2) $\frac{Q}{4\pi\varepsilon_0 r_1^2}$

(3) $\frac{Q r_1^2}{4\pi\varepsilon_0 R^4}$

(4) $\frac{Q r_1^2}{3\pi\varepsilon_0 R^4}$

28) Charge Q is distributed to two different metallic spheres having radius R and 2R such that both spheres have equal surface charge density, then charge on large sphere is :-

(1) $\frac{4Q}{5}$

(2) $\frac{Q}{5}$

(3) $\frac{3Q}{5}$

(4) $\frac{5Q}{4}$

29) A point charge q is placed inside a conducting spherical shell of inner radius 2R and outer radius 3R at a distance of R from the centre of shell. The electric potential at the centre of shell will be $\frac{1}{4\pi\varepsilon_0}$ times of :-

(1) $\frac{q}{2R}$

(2) $\frac{4q}{3R}$

(3) $\frac{5q}{6R}$

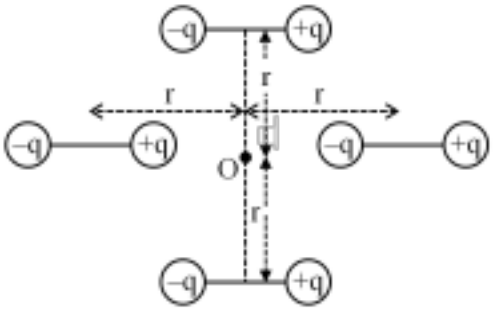
(4) $\frac{2q}{3R}$

30)

Net electric field at the given point O, due to all the four identical short electric dipoles (of dipole moment p each) shown in the figure, is $\left(k = \frac{1}{4\pi\varepsilon_0}\right)$

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(1) $\frac{kp}{r^3}$

(2) $\frac{2kp}{r^3}$

(3) $\frac{kp}{2r^3}$

(4) $\frac{4kp}{r^3}$

31) A parallel plate capacitor is connected with a battery whose potential is constant . If the plates of capacitor are shifted apart, then the intensity of electric field

(1) decreases and charge on plates also decreases.

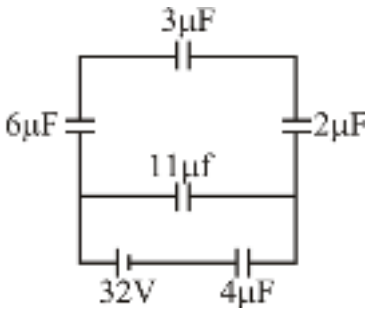
(2) remains constant but charge on plates decreases.

(3) remains constant but charge on plates increases.

(4) increases but charge on plates decreases.

32)

Choose correct option regarding given circuit.



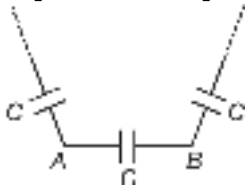
(1) Potential difference across 4μF is 20V.

(2) Charge at 3μF is 8μC.

(3) Energy stored in 2μF is 32μJ.

(4) Ratio of charges on 3μF & 11μF are 3 : 11.

33) On each side of a polygon of n sides a capacitor of capacitance C is placed as shown in figure.

Equivalent capacitance across A and B is 

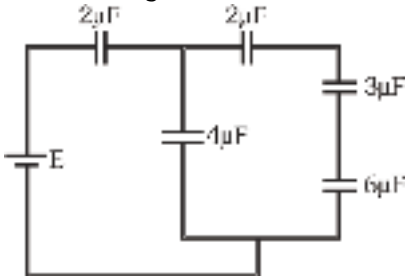
(1) $\frac{(n-1)C}{n}$

(2) $\frac{nC}{n-1}$

(3) (n – 1)C

(4) nC

34) In the given circuit, if charge on 6μF capacitor is 10 μC, then charge on 4 μF capacitor will be



(1) 40 μC

(2) 10 μC

(3) 20 μC

(4) 30 μC

35) The equivalent capacitance of three capacitors of capacitance C_1, C_2 and C_3 connected in parallel, is 12 units and the product $C_1 C_2 C_3 = 48$. When the capacitance C_1 and C_2 are connected in parallel, the equivalent capacitance is 6 units. Then the capacitances are:-

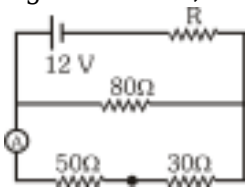
(1) 1.5, 2.5, 8

(2) 2, 3, 7

(3) 2, 4, 6

(4) 1, 5, 6

36) In the given circuit, the ammeter A assumed to have negligible resistance, reads 0.1 A. The value of R is :-



(1) 6Ω

(2) 8Ω

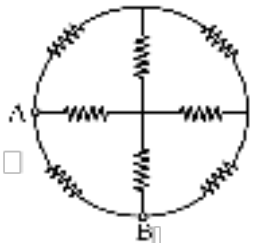
(3) 16Ω

(4) 20Ω

37) Eight resistances each of resistance 5Ω are connected in the circuit as shown in figure. The

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equivalent resistance between A & B is–

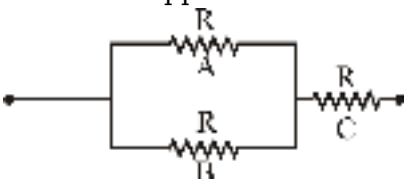
(1) $\frac{8}{3}\Omega$

(2) $\frac{16}{3}\Omega$

(3) $\frac{15}{7}\Omega$

(4) $\frac{19}{2}\Omega$

38) The three identical resistance A, B and C each of R are connected as shown in figure. When some potential difference is applied across the network, the thermal powers dissipated by A, B and C



are in the ratio :-

(1) 1 : 1 : 2

(2) 4 : 4 : 1

(3) 1 : 1 : 4

(4) None of these

39) Nichrome and manganin is widely used in wire wound standard resistors, this is because of their:

(1) Temperature independent resistivity

(2) Strong dependency of resistivity on temperature

(3) Mechanical strength

(4) Very weak temperature dependent resistivity

40) In a meter bridge experiment, the ratio of the left gap resistance to right gap resistance is 2 : 3, the balance point from left is -

(1) 60 cm

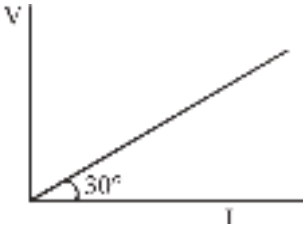
(2) 50 cm

(3) 40 cm

(4) 20 cm

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41) Find the value of resistance for V-I graph :-



(1) $\sqrt{3}\ \Omega$

(2) $\frac{1}{\sqrt{3}}\Omega$

(3) 2 Ω

(4) 4 Ω

42) Two cells of equal e.m.f. and of internal resistances r_1 and r_2 ($r_1 > r_2$) are connected in series. On connecting this combination to an external resistance R, it is observed that the potential difference across the first cell becomes zero. The value of R will be

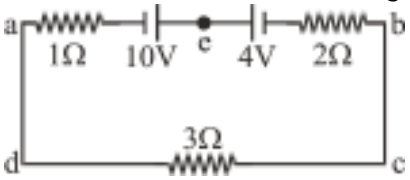
(1) $r_1 + r_2$

(2) $r_1 - r_2$

(3) $\frac{r_1 + r_2}{2}$

(4) $\frac{r_1 - r_2}{2}$

43) In the circuit shown in figure, the magnitude of the flow of current, would be :-



(1) $\frac{5}{3}$ A

(2) $\frac{7}{3}$ A

(3) 1 A

(4) 2 A

44) Which of the following statement is True regarding Null point in finding resistance of unknown wire using Meter Bridge.
(i) At Null Point, current through galvanometer is zero.
(ii) At Null Point, condition is balanced Wheat stone bridge.
(iii) Deflection of galvanometer in same direction, either side of Null point.

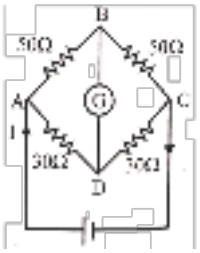
(1) (i) only

(2) (i) & (iii)

(3) (i) & (ii)

(4) All

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45) In the given circuit, the current in the arm BD is

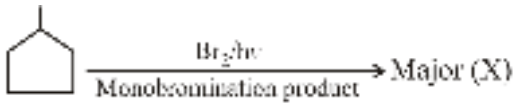
(1) I/2

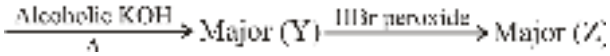
(2) zero

(3) 2I

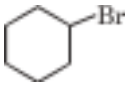
(4) I/3

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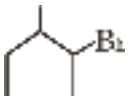


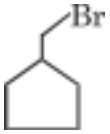
1) 

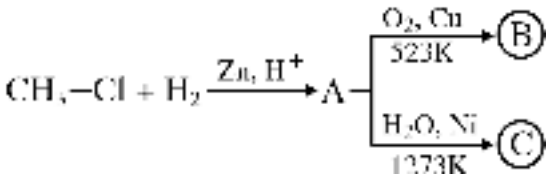
Product (Z) is :

(1) 

(2) 

(3) 

(4) 

2) 

Here Ⓑ & Ⓒ are respectively

(1) CO_2, CH_3OH

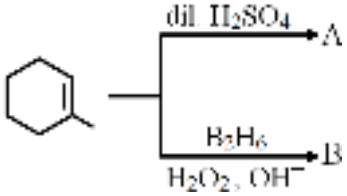
(2) CH_3OH, $CO + 3H_2$

(3) HCHO, $CO + 3H_2$

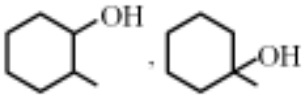
(4) CO_2, HCOOH

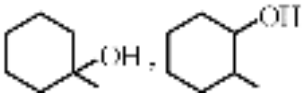
3)

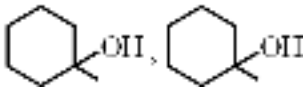
Consider

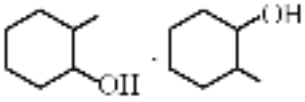


Here A and B will be respectively?

(1) 

(2) 

(3) 

(4) 

4) The compound with the highest boiling point is–

(1) n-hexane

(2) n-pentane

(3) 2,2-dimethyl propane

(4) propane

5) 'A' and 'B' respectively are :

A $\xrightarrow[(2)Zn-H_2O]{(1)O_3}$ Butanedial + Glyoxal

B $\xrightarrow[(2)Zn-H_2O]{(1)O_3}$ 5-oxohexanal

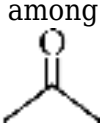
(1) 1-methylcyclohex-1, 3-diene & cyclopentene

(2) Cyclohex-1, 3-diene & cyclopentene

(3) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene

(4) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

6) Identify total number of electrophile among following species ?

CO_2 , SO_3, I^- , Br^- , N_3^- , Me–C(=O)–H ,  , Me–C(=O)–Cl , $CH_3^{\oplus}$

(1) 4

(2) 5

(3) 6

(4) 3

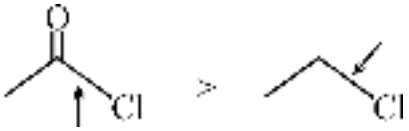
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7) Incorrect order(s) for bond lengths of indicated bonds :

(A) 

(B) 

(C) 

(D) 

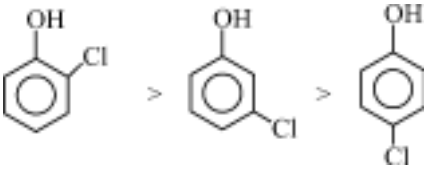
(1) Only A, B

(2) Only A, B, C

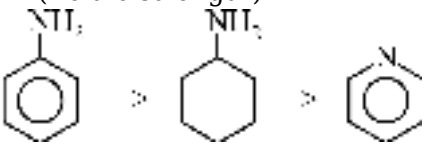
(3) Only C, D

(4) Only B, C

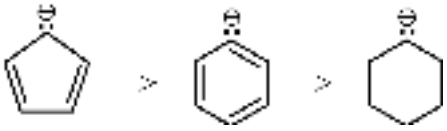
8) Which is/are correct :

(A) 

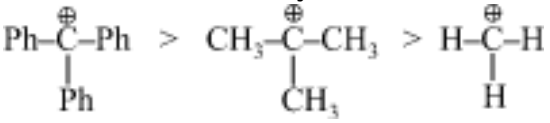
(Acidic strength)

(B) 

(Basic strength)

(C) 

(Carbanion stability)

(D) 

(Carbocation stability)

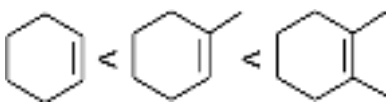
(1) Only A, D

(2) Only B, D

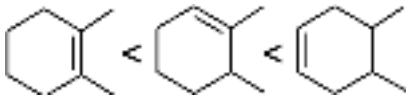
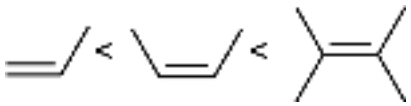
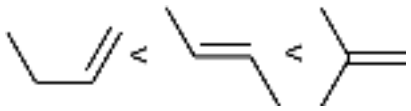
(3) Only C, D

(4) Only A, C, D

9)

Column - I		Column - II	
(P)		(1)	Correct order of stability of alkene

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(Q)		(2)	Correct order of heat of combustion
(R)		(3)	Correct order of heat of hydrogenation
(S)		(4)	Correct order of C=C bond length

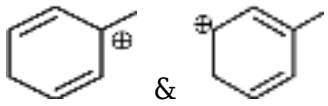
(1) P→1,4; Q→1,2,4; R→2,3; S→1,2,4

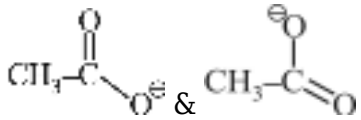
(2) P→1,2,4; Q→2,3; R→1,2,4; S→1,4

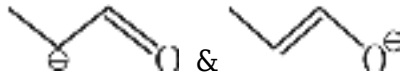
(3) P→1,2; Q→1,4; R→2,3; S→1,2,4

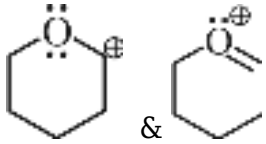
(4) P→1,4; Q→1,2,4; R→2,3; S→1,2,4

10) In which of the following molecules first resonating structure is more stable than second.

(1) 

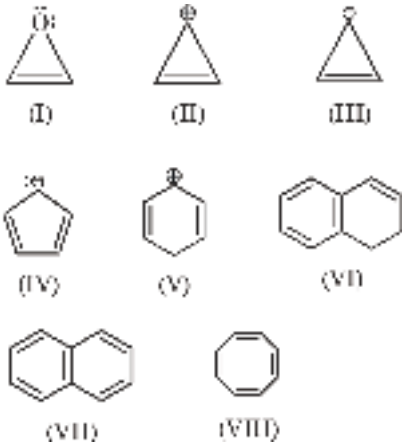
(2) 

(3) 

(4) 

11)

Number of aromatic species are :



(1) 3

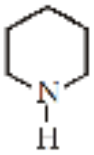
(2) 4

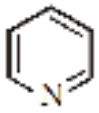
(3) 5

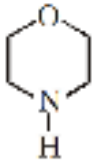
(4) 6

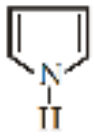
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12) In the following compounds :

(i) 

(ii) 

(iii) 

(iv) 

Order of basicity is :

(1) iii > i > ii > iv

(2) ii > iii > i > iv

(3) iv > iii > ii > i

(4) i > iii > ii > iv

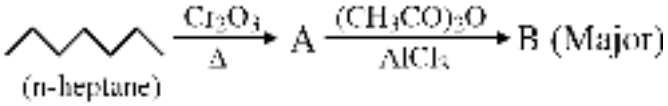
13) In detection of 'P' by fusing with Na_2O_2 and then extracting it with water, we add HNO_3(conc.) and ammonium molybdate to get

(1) Yellow ppt of $(NH_3)_3PO_4$

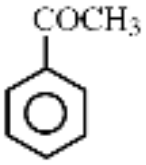
(2) Yellow ppt of $(NH_4)_3PO_4.12MoO_3$

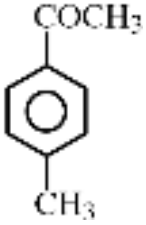
(3) White ppt of $(NH_4)_3PO_4$

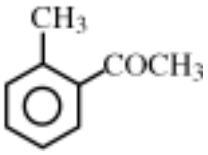
(4) White ppt. of $(NH_4)_2HPO_3$

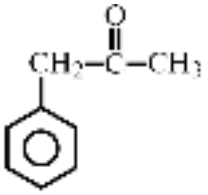
14) 

Here B is

(1) 

(2) 

(3) 

(4) 

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15) Match **List-I** with **List-II** :

	List-I		**List-II**
(a)	Separation of aniline-water mixture	(i)	Fractional distillation
(b)	Separation of aniline-chloroform mixture	(ii)	Distillation under reduced pressure
(c)	Separation of glycerol from spent-lye	(iii)	Simple distillation
(d)	Separation of different fractions of crude oil	(iv)	Steam distillation

Choose the **correct answer** from the options given below :

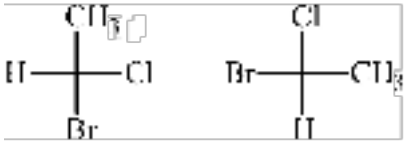
(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

16) The following structures represent a pair of :-



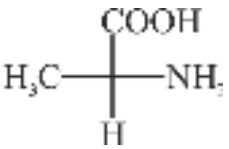
(1) Enantiomer

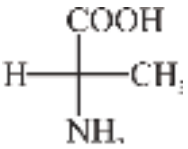
(2) Diastereomer

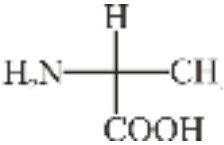
(3) Meso compound

(4) Homomers

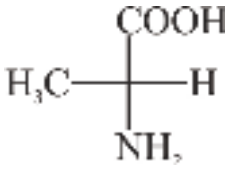
17) Among the following amino acids, R-enantiomer is:

(1) 

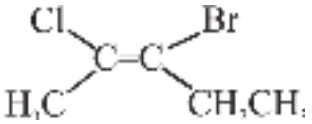
(2) 

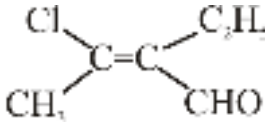
(3) 

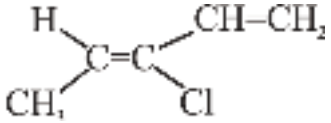
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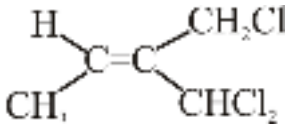
(4) 

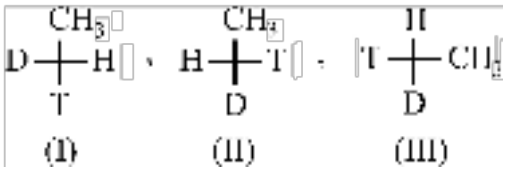
18) Which of the following represent E-configuration :-

(1) 

(2) 

(3) 

(4) 

19) 

Choose the correct statement

(1) I & II have R configuration

(2) I & III are enantiomers

(3) II & III are enantiomers

(4) all are identical

20) How many structural isomers of alcohol with the formula C_4H_9OH are possible ?

(1) 4

(2) 3

(3) 5

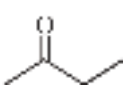
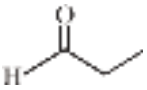
(4) 6

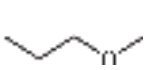
21)

Which is correctly matched ?

(1)  and  Position isomers

(2)  and 
Functional group isomers

(3)  and  Functional isomers

(4)  and  Chain isomers

22) 0.1 gm of organic compound was analysed by Kjeldahl's method. In analysis produced NH_3 absorbed in 25 ml N/5 H_2SO_4. The remaining acid required 20 ml N/10 NaOH for neutralisation. Calculate percentage of nitrogen in organic compound :-

(1) 28 %

(2) 56 %

(3) 42 %

(4) 70 %

23) At 300 K the half-life of a sample of a gaseous compound initially at 1 atm is 100 sec. When the pressure is 0.5 atm the half-life is 50 sec. The order of reaction is :

(1) 0

(2) 1

(3) 2

(4) 3

24) A first order reaction has half life 12 hours. What percentage of the reactant will remain after 24 hours :-

(1) 75%

(2) 25%

(3) 87.5%

(4) 93.75%

25) For which of the following reactions, the temperature coefficient (μ) is maximum?

(1) $A \rightarrow B$ $E_a = 50$ kJ

(2) $P \rightarrow Q$ $E_a = 40$ kJ

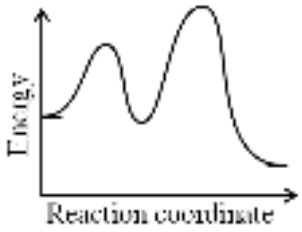
(3) $X \rightarrow Y$ $E_a = 60$ kJ

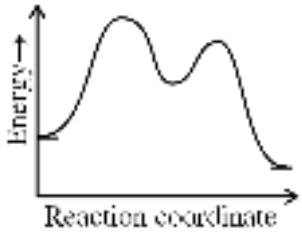
(4) $W \rightarrow Z$ $E_a = 80$ kJ

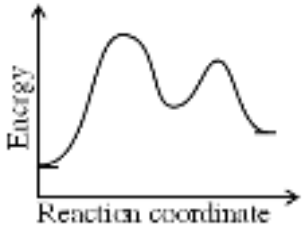
26) Reactant R converts to product P through given mechanism (with net absorption of heat)
$R \rightarrow X$; slow ;
$X \rightarrow P$; fast ;
which of the following represent the above reaction mechanism :-

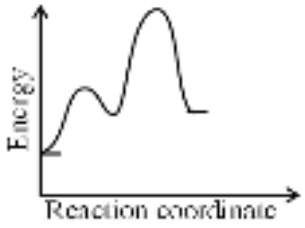
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(1) 

(2) 

(3) 

(4) 

27) For the reaction 2A + B → C, the values of initial rate at different reactant concentrations are given in the table below. The rate law for the reaction is :

[A] (mol L^{-1})	[B] (mol L^{-1})	Initial Rate (mol $L^{-1}s^{-1}$)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

(1) Rate = k [A][B]

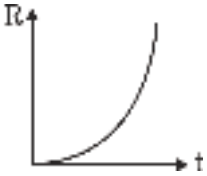
(2) Rate = k $[A]^2[B]^2$

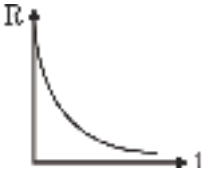
(3) Rate = k $[A][B]^2$

(4) Rate = k $[A]^2[B]$

28) If decomposition reaction A(g) → B(g) follows first order kinetics then the graph of rate of consumption (R) of A against time t will be :-

(1) 

(2) 

(3) 

(4) 

29) ***Assertion* (A) :-** Vapour pressure of 0.5 M sugar solution is more than 0.5 M KCl solution.
***Reason* (R) :-** The relative lowering of vapour pressure is directly proportional to the number of solute particles present in solution.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

30) An aqueous solution freezes at –0.186°C (K_f =1.86 K Kg mol^{-1} ; K_b = 0.512 K Kg mol^{-1}). What is the elevation in boiling point of that solution ?

(1) 0.186 °C

(2) 0.512 °C

(3) $\frac{0.512}{1.86}$ °C

(4) 0.0512 °C

31) The freezing point of a solution containing 0.2 g of acetic acid in 20.0 g benzene is lowered by 0.45°C. Calculate the degree of dimerization of acetic acid in benzene. K_f for benzene is 5.12 K mol^{-1} Kg :-

(1) 0.527

(2) 0.80

(3) 0.945

(4) None of these

32) Two liquid A and B have vapour pressure in the ratio $P_A^0 : P_B^0 = 1 : 3$ at a certain temperature. If the ratio of mole fractions of A to B in the vapour phase is 4 : 3, then the mole fraction of B in the solution at the same temperature is :

(1) $\frac{1}{5}$

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(2) $\frac{2}{3}$

(3) $\frac{4}{5}$

(4) $\frac{1}{4}$

33) Calculate freezing point of 0.2 molal aq. solution of a mono protic acid HX(α = 20%)
(k_f for water = 1.86 KKg/mol)

(1) 273.4464

(2) 272.5536K

(3) 262.15K

(4) 270.152K

34) Ratio of solubilities of gases N_2 & O_2 in water (as mole of gas in large volume of water) from air at 25°C & 1 atm will be, if air is 20 % by mole of O_2 & 80% by mole of N_2
Given : K_H (N_2) = 2×10^4 atm
K_H (O_2) = 10^4 atm

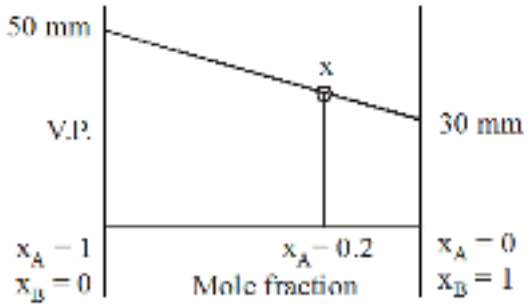
(1) 8 : 1

(2) 1 : 8

(3) 2 : 1

(4) 1 : 2

35) The variation of total vapour pressure of an ideal solution of two liquids A and B is depicted aside. The total vapour pressure denoted by the point 'X' will be:



(1) 46 mm

(2) 40 mm

(3) 34 mm

(4) 32 mm

36) What will be the EMF for the given cell ?
$Pt(|H_2(g),P_1)|H^+(aq,1M)||H^+(aq.,1M)|(H_2(g),P_2)|Pt(s)$

(1) $\frac{RT}{F}\ln\frac{P_1}{P_2}$

(2) $\frac{RT}{2F}\ln\frac{P_1}{P_2}$

(3) $\frac{RT}{F}\ln\frac{P_2}{P_1}$

(4) $\frac{RT}{2F}\ln\frac{P_2}{P_1}$

37) How many coulombs of electricity are required for the reduction of 1 mole of MnO_4^- to Mn^{2+} ?

(1) 96500 C

(2) 9.65×10^6 C

(3) 4.83×10^5 C

(4) 1.93×10^5 C

38) (i) $Cu + 2HCl \rightarrow CuCl_2 + H_2(g)$;
$E^o_{Cu^{2+}/Cu} = +0.34$ V

(ii) $Zn + 2HCl \rightarrow ZnCl_2 + H_2(g)$;
$E^o_{Zn^{2+}/Zn} = -0.76$ V

(iii) $Ag + 2HCl \rightarrow AgCl + H_2(g)$;
$E^o_{Ag^+/Ag} = +0.80$ V

Which of the above reactions is feasible ?

(1) (ii)

(2) (i)

(3) (iii)

(4) All of the above

39)

A 5 ampere current is passed through a solution of zinc sulphate for 40 minutes. Find the amount of zinc deposited at cathode (atomic weight Zn = 65.38) :-

(1) 40.65 gm

(2) 4.065 gm

(3) 0.4065 gm

(4) 65.04 gm

40) The ratio of weights of hydrogen gas liberated and magnesium deposited by passing the same amount of electricity from aqueous H_2SO_4 and fused $MgSO_4$ are respectively :

(1) 1 : 8

(2) 1 : 12

(3) 1 : 16

(4) None

41) A 1.0 M solution of Cd^{2+} is added to excess iron and the system is allowed to reach equilibrium. What is the concentration of Cd^{2+} at equilibrium?

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$Cd^{2+}(aq) + Fe(s) \rightleftharpoons Cd(s) + Fe^{2+}(aq)$; $E° = 0.037$ V

(1) 0.195 M
(2) 0.097 M
(3) 0.053 M
(4) 0.145 M

42) How many faraday charge is required to convert one mole of C_5H_{12} in to CO_2 ?

(1) 8
(2) 12
(3) 20
(4) 32

43) If the E^0_{cell} for a reaction has a positive value, then which of the following gives the correct relationship for the values of $\Delta G°$ and K_{eq} ?

(1) $\Delta G° > 1$; $K_{eq} < 1$
(2) $\Delta G° < 1$; $K_{eq} > 0$
(3) $\Delta G° > 0$; $K_{eq} > 1$
(4) $\Delta G° < 0$; $K_{eq} > 1$

44) Suitable indicator for the acid-base titration of HCOOH against KOH is.

(1) Methyl orange
(2) Methyl red
(3) Phenolphthalein
(4) Thymol Blue

45) The enthalpies of neutralization of a weak base AOH and a strong base BOH by HCl are –12250 cal/mol and –13000 cal/mol respectively. When one mole of HCl is added to a solution containing 1 mole of AOH and 1 mole of BOH, the enthalpy change was –12500 cal/mol. In what mole ratio the acid neutralise base AOH and BOH respectively ?

(1) 2 : 1
(2) 2 : 3
(3) 1 : 2
(4) 3 : 2

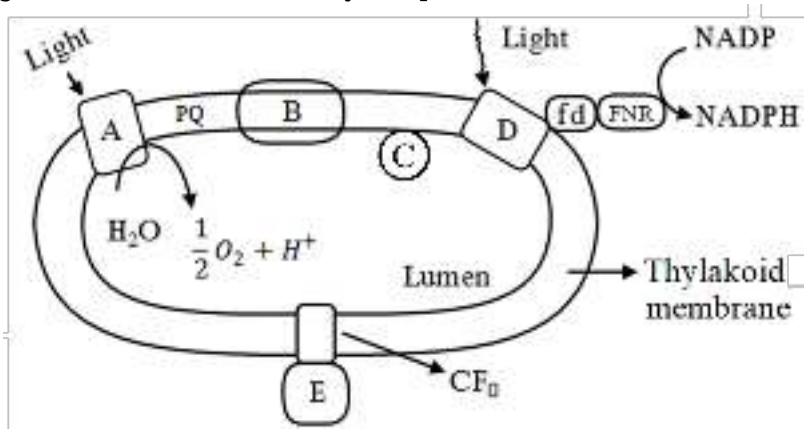
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1) Which statements are incorrect w.r.t. Calvin cycle ?
(A) Carboxylation is the fixation of CO_2 into a stable organic intermediate.
(B) Regeneration of the CO_2 acceptor molecule RuBP is crucial if the cycle is to continue uninterrupted.
(C) To make one molecule of sucrose 6 turns of the cycle are required.

(D) For every CO_2 molecule entering the Calvin cycle, 3 molecules of NADPH and 2 of ATP are required.
(E) Regeneration is the most crucial step of the Calvin cycle where CO_2 is utilised for the carboxylation of RuBP.

(1) C, D and E only
(2) D and E only
(3) A, D and E only
(4) B, D and E only

2) Given below is the diagrammatic sketch. Identify the parts labelled A,B,C,D and E and select the right option about them.



(1) A - PS - I; B - PS - II; C - Plastocyanin; D - Cytochrome; E - CF_1
(2) A - PS - II; B - PS - I; C - Plastocyanin; D - Cytochrome; E - CF_1
(3) A - PS - II; B - Cytochrome; C - CF_1; D - PS - I; E - Plastocyanin
(4) A - PS - II; B - Cytochrome; C - Plastocyanin; D - PS - I; E - CF_1

3) C_4 plants are special. They have special type of (i), they tolerate (ii), they show response to (iii), they lack a process called (iv) and have (v) productivity of Biomass.

	(i)	(ii)	(iii)	(iv)	(v)
(1)	Leaf morphology	Low temperature	Low light intensity	Photorespiration	Lesser
(2)	Leaf anatomy	High temperature	High light intensity	Photorespiration	Greater
(3)	Stem anatomy	Low temperature	Low light intensity	Photosynthesis	Lesser
(4)	Stem anatomy	High temperature	High light intensity	Photorespiration	Greater

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(1) 1

(2) 2

(3) 3

(4) 4

4) If we remove accessory pigments from plant then what will be the effect ?

(1) The efficiency of photosynthesis will increase because all the light energy become available to chlorophyll 'a'

(2) Range of wavelengths, which are utilised in photosynthesis, become narrower

(3) Due to less amount of light energy the possibility of photo-oxidation of chlorophyll 'a' will decrease

(4) The photolysis of water and formation of ATP will not occur

5) **Assertion :** Immediately after light becomes unavailable, the biosynthesis process continues for some time, and then stops.
Reason : Biosynthetic phase of photosynthesis does not directly depend on light but is dependent on the products of light reaction.

(1) Both Assertion & Reason are True & Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion.

(3) Assertion is True but Reason is False.

(4) Both Assertion & Reason are False.

6) The thylakoid membrane bears several F_0 – F_1 particle (ATP synthase). Which of following is incorrect for these particles?

(1) One of its part (F_0) is embedded in the membrane and forms transmembrane channel that carries out facilitated diffusion of protons across the membrane.

(2) Its other part (F_1) protrudes out facing towards the stroma

(3) The catalytic sites for ATP formation are located in F_1 part

(4) F_1 is intrinsic protein

7) Match the column-I with column-II and choose the correct combination from the options given below.

	Column-I		**Column-II**
(A)	Oxygen evolving complex	(I)	Pigments
(B)	Proton gradient	(II)	High oxygen concentration
(C)	Absorb light at specific wavelength	(III)	ATP synthesis
(D)	Photorespiration	(IV)	Photolysis of H_2O

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

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(2) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

8) Chlorophyll is :-

(1) Iron-porphyrin with hydrophilic head and hydrophobic tail

(2) Mg porphyrin with hydrophobic head and hydrophilic tail

(3) Mg porphyrin with hydrophilic head and hydrophobic tail

(4) Iron porphyrin with hydrophobic head and hydrophilic tail.

9) Light reaction provides:

(1) O_2, H-acceptor and photons

(2) H-donor, O_2 and glucose

(3) H-donor, O_2 and ATP

(4) Electron acceptor, H-acceptor and CO_2

10) C_4 plants consume _____ molecules of $NADPH_2$ to fix 4 CO_2, while C_3 plants require _____ molecules of $NADPH_2$ for fixation of same amount of CO_2 :-

(1) 8, 8

(2) 4, 4

(3) 6, 6

(4) 8, 12

11) Match the column I, II and III with respect to Kreb's cycle and choose the correct option :-

Column-I (Reaction)		**Column-II (Enzyme)**		**Column-III (Energy equivalent process)**	
(A)	Isocitrate → α-KGA	(I)	Succinate dehydrogenase	(1)	Oxidative decarboxylation
(B)	Succinyl Co-A → Succinic acid	(II)	Isocitrate dehydrogenase	(2)	SLP
(C)	Succinic acid → Fumaric acid	(III)	Thiokinase	(3)	$NADH.H^+$ formation & decarboxylation
(D)	α-KGA → Succinyl CoA	(IV)	Dehydrogenase complex	(4)	$FADH_2$ formation

(1) A -II - 3, B - III - 2, C - I - 4, D - IV - 1

(2) A -I - 1, B - II - 2, C - III - 3, D - IV - 4

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(3) A -III - 2, B - II - 3, C - IV - 4, D - I - 1

(4) A -IV - 4, B - III - 3, C - II - 2, D - I - 1

12) During complete oxidation of pyruvate which of the following ratios are correct regarding release of $NADH + H^+$, $FADH_2$ & CO_2 ?

(1) $NADH + H^+ : FADH_2 : CO_2 :: 2 : 1 : 4$

(2) $NADH + H^+ : FADH_2 : CO_2 :: 4 : 3 : 2$

(3) $NADH + H^+ : FADH_2 : CO_2 :: 4 : 1 : 3$

(4) $NADH + H^+ : FADH_2 : CO_2 :: 3 : 1 : 3$

13) Read the following statements :
(a) Glycolysis is common both to aerobic and anerobic respiration.
(b) Glycolysis involves complete oxidation of glucose.
(c) ATP is utilised in three different steps of glycolysis.
(d) Energy is released by reduction of respiratory substrates.
Which of the above statements are not correct.
Options :-

(1) Only (a), (b) and (c)

(2) Only (b), (c) and (d)

(3) Only (a), (c) and (d)

(4) (a), (b), (c) and (d)

14) Match the column-A & column-B to find out the correct answer :

	Column-A		**Column-B**
(i)	Chain of Ten reactions	(a)	Invertase
(ii)	Conversion of sucrose into glucose & fructose	(b)	DHAP & PGAL
(iii)	Hexokinase	(c)	Glycolysis
(iv)	Fructose 1, 6 Bisphosphate	(d)	Conversion of glucose in to glucose-6-phosphate

(1) i-a, ii-b, iii-c, iv-d

(2) i-b, ii-c, iii-d, iv-a

(3) i-c, ii-a, iii-d, iv-b

(4) i-d, ii-a, iii-b, iv-c

15) Pyruvic acid, the key product of glycolysis can have many metabolic fates. Under aerobic condition it forms:-

(1) lactic acid

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(2) Glucose + CO_2

(3) acetyl CoA + CO_2

(4) ethanol + CO_2

16) Which of the following component of fat is synthesized from dihydroxy acetone phosphate ?

(1) Fatty acid

(2) Glycerol

(3) Both (1) and (2)

(4) Cholesterol

17) **Statement-I :** Auxin was isolated by Charles Darwin and his son Francis Darwin when they observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism)
Statement-II : The bakane (foolish seedling) disease of rice seedling was caused by a fungal pathogen *Gibberella Fujikuroi*. E. Kurosawa reported the appearance of symptoms of the disease in rice seedlings when they were treated with sterile filtrates of the fungus.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

18) **Statement-I :** There are more than 100 gibberellins reported from widely different organisms such as fungi and higher plants.
Statement-II : The ability of GAs to cause an increase in length of axis is used to increase the length of grapes stalks.

(1) Only statement I is correct.

(2) Only statement II is correct.

(3) Both statement I and II are correct.

(4) Both statement I and II are incorrect.

19) Match List-I with List-II :-

List-I		**List-II**	
(A)	Widely used as herbicides	(I)	Cytokinins
(B)	Overcome the apical dominance	(II)	Ethylene
(C)	Petiole elongation in deep water rice plants	(III)	ABA
(D)	Inhibits seed germination	(IV)	2, 4-D

Choose the option with all correct matches :-

(1) (A)-III, (B) - I, (C) - II, (D) - IV

(2) (A)-III, (B) - IV, (C) - I, (D) - II

(3) (A)-IV, (B) - III, (C) - I, (D) - II

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(4) (A)-IV, (B) - I, (C) - II, (D) - III

20) Arrange the following events in the correct sequence for development of a typical plant:
A. Plantlet B. Seedling
C. Mature plant D. Senescence

(1) B → A → C → D
(2) A → B → C → D
(3) A → C → B → D
(4) B → C → D → A

21) What kind of a curve can you expect in a tree showing seasonal activities?

(1) Linear
(2) Hyperbola
(3) Sigmoid
(4) J - Shaped

22) Thinning of cotton, cherry and walnut is induced by-

(1) Ethylene
(2) Auxin
(3) Cytokinin
(4) Abscisic acid

23) Fruits in mango and coconut are developed from :

(1) Multicarpellary gynoecium with inferior ovary
(2) Monocarpellary gynoecium with superior ovary
(3) Multicarpellary gynoecium with superior ovary
(4) Monocarpellary gynoecium with inferior ovary

24) Axillary bud is present in axil of :

(1) Simple leaf only
(2) Pinnately compound leaf only
(3) Palmately compound leaf only
(4) Simple and compound leaf both

25) Epipetalous androecium and axile placentation are present in _____ .

(1) Musturd
(2) Brinjal
(3) Pea
(4) Lilly

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26) Given below are two statements one is labelled as Assertion (A) and other is labelled is Reason (R)
Assertion (A) : Within the seed coat is the embryo consisting of embryonal axis and one or two cotyledons.
Reason (R) : The cotyledon is fleshy and full of reserve food materials in dicot plant.
In the light of above statements choose correct answer from the options given below :

(1) Both A and R are true and R is correct explanation of A
(2) Both A and R are true and R is not correct explanation of A
(3) A is true and R is false
(4) A is false and R is true

27) **Statement-I** :- In *Calotropis* valvate aestivation is present.
Statement-II :- In *Cassia* and gulmohar the aestivation is vexillary.
In the light of above statements choose the correct answer from the following option :-

(1) Statement I is correct and II is incorrect
(2) Both statement are correct
(3) Statement I is incorrect and II is correct
(4) Both statement are in correct

28) How many of the following have axile placentation of ovule in ovary?
Pea, Bean, Tomato, Lily, Lemon, Mustard, *Primrose*, Sunflower

(1) Three
(2) Four
(3) Five
(4) Six

29) Select the incorrect statements from given below :
(i) Actinomorphic flowers can be seen in mustard and china rose.
(ii) In Pea flowers are zygomorphic and incomplete.
(iii) In *Salvia* and musturd- anthers are arrange on different hight due to variation in the length of filaments.
(iv) Apocarpus, multicarpellary gynoecium present in lotus.
(v) The stamens may be united in to two bundles in citrus.

(1) (ii), (iii), (iv)
(2) (ii), (iv), (v)
(3) (ii), (v)
(4) (i), (iii), (v)

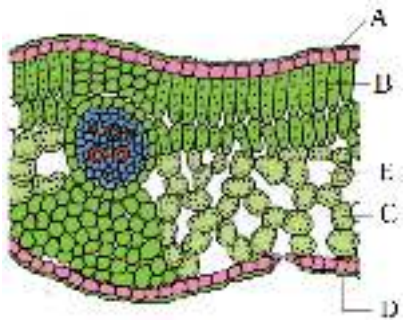
30) Read the following statements and choose correct options -
A. Epidermal cells are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole.
B. Ground tissue systems consists of simple tissues such as parenchyma, collenchyma and sclerenchyma
C. The pith is small or inconspicuous in monocot root

D. The cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath in dicot stem
E. The veins vary in thickness in the reticulate venation of the dicot leaves.

(1) A , B and D are incorrect
(2) A, C and E are incorrect
(3) Only C is incorrect
(4) All are correct

31) **Statement-I :** The Parenchymatous cells which lie between xylem and phloem in roots is called conjuctive Tissue.
Statement-II : Initiation of lateral roots and vascular cambium during the secondary growth takes place in cell of pericycle.

(1) Statement-I and statement-II both are correct
(2) Statement-I and statement-II both are incorrect
(3) Only statement-I is correct
(4) Only statement-II is correct



32) Identify A, B, C, D and E in figure given below

	A	B	C	D	E
1	Adaxial epidermis	Spongy mesophyll	Palisade mesophyll	Abaxial epidermis	Air cavity
2	Abaxial epidermis	Palisade mesophyll	Spongy mesophyll	Adaxial epidermis	Air cavity
3	Adaxial epidermis	Palisade mesophyll	Spongy mesophyll	Abaxial epidermis	Air cavity
4	Abaxial epidermis	Spongy mesophyll	Palisade mesophyll	Adaxial epidermis	Air cavity

(1) 1
(2) 2
(3) 3
(4) 4

33) Identify the set of correct statements :-
(A) Apical meristem occupies the tip of shoot only.
(B) Intercalary meristem occurs between mature tissues.
(C) Parenchyma cells are generally isodiametric.

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(D) The collenchyma occurs in layer below the epidermis in monocot stem.
(E) Sclerenchyma provides mechanical support to plant organs.
Choose the correct answer from the options given below :-

(1) (A), (B) & (E) only
(2) (B), (C) and (E) only
(3) (A), (C) and (E) only
(4) (B), (D) and (E) only

34) How many in the list given below are the part of ground tissue system :
Epidermis, Root hairs, Xylem, Phloem, Cortex, Pericycle, Pith, Endodermis.

(1) One
(2) Two
(3) Three
(4) Four

35) In dorsiventral leaves, the location of xylem and spongy tissue towards :

(1) Abaxial surface
(2) Adaxial surface
(3) Adaxial and abaxial surface respectively
(4) Abaxial and adaxial surface respectively

36) Choose the incorrect pair :

(1) Cuticle is absent - Roots
(2) Primary function of epidermis - Protection
(3) Regulate transpiration - Stomata
(4) Bean shape stomatal cells - Subsidary cells

37) Arrange the following events in correct sequence for the development of male gametophyte in angiosperms :
(A) Formation of microspore mother cells.
(B) Meiosis to form microspore tetrads.
(C) Pollen grain with two male gametes.
(D) Pollen grain with generative and vegetative cells.
(E) Formation of microsporangium
Options :

(1) E → B → A → D → C
(2) E → A → B → D → C
(3) A → B → C → D → E
(4) D → C → E → B → A

38) **Statement-I:** In many citrus and mango varieties, polyembryony is observed where some nucellar cells near the embryo sac begin to divide and differentiate into multiple embryos, resulting

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in the presence of more than one embryo in a single seed.
Statement-II: All the embryos formed in polyembryony are a result of fertilisation and are therefore genetically different from the parent plant.

(1) Both statements are correct.
(2) Both statements are incorrect.
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct.

39) Select the incorrect statement :-

(1) The proximal end of the filament is attached to the thalamus or the petal of the flower.
(2) A typical angiosperm anther is bilobed & dithecous.
(3) Sporogenous tissue undergo mitotic division to form microspore.
(4) Sporogenous tissue occupies the centre of each microsporangium.

40) Match the column-I with column-II and select the correct answer.

Column-I		**Column-II**	
(A)	Tapetum wall	(i)	25-50 µm in diameter
(B)	Pollen grains	(ii)	Sporopollenin
(C)	Exine layer	(iii)	Nourishes the developing pollen grain
(D)	Sporogenous tissue	(iv)	Centre of microsporangium

(1) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
(2) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)
(3) (A)-(i), (B)-(ii), (C)-(iv), (D)-(iii)
(4) (A)-(iv), (B)-(i), (C)-(iii), (D)-(ii)

41) **Assertion (A) :** Triple fusion leads to the formation of primary endosperm nucleus.
Reason (R) : Triple fusion involves fusion of one male gamete with the egg cell.

(1) Both Assertion & Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion & Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

42) The cells of which of the following layers of anther wall are multinucleated & possess dense cytoplasm ?

(1) Middle layer
(2) Tapetum
(3) Epidermis
(4) Endothecium

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43) In which part of the flower are megasporangia typically located ?

(1) Anther
(2) Stigma
(3) Style
(4) Ovary

44) Often flowers of animal pollinated plants are specifically adapted for :-

(1) All species of animals
(2) All species of insects
(3) Particular species of animals
(4) Particular species of plants

45) Which of the following statement is correct ?

(1) Seed is often described as a fertilised ovary.
(2) Perisperm is a part of the seed in black pepper and beet
(3) All monocot seeds are albuminous.
(4) Non-endospermic seeds have both residual endosperm and residual nucellus.

46) Read the following terminology of column I & II and identify the correct match :

	Column I		**Column II**
(1)	Neurohypophysis	(a)	TSH
(2)	Adenohypophysis	(b)	Oxytocin
(3)	CNS	(c)	Vasopressin
(4)	Neuron	(d)	Brain and spinal cord
		(e)	Cell body, dendrites and axon
		(f)	ACTH

(1) 1-a,f; 2-b,f; 3-e,f; 4-d
(2) 1-b,c; 2-a,f; 3-d; 4-e
(3) 1-b,c; 2-a,f; 3-e; 4-d
(4) 1-e; 2-b,c; 3-a,f; 4-d

47) Choose the correct option from given below :
(a) Association areas are clearly sensory.
(b) Association areas are neither clearly sensory nor motor.
(c) Association areas are responsible for complex functions.
(d) Association areas are clearly motor.

(1) a, c
(2) a, b, c
(3) b, c

(4) a, b, c, d

48) Which of the following statements is incorrect ?

(1) Pons contains fibre tracts
(2) Medulla oblongata controls reflexes
(3) Cerebellum has a convoluted surface
(4) Brainstem is made up of cerebellum, pons and midbrain

49) How many of the following statements are correct ?
(i) The brain stem is made up of midbrain, pons and medulla oblongata.
(ii) Cerebellum connects the brain to the spinal cord.
(iii) Medulla oblongata controls gastric secretion and cardiovascular reflexes.
(iv) Pons helps in connecting different regions of the brain.

(1) One
(2) Two
(3) Three
(4) Four

50) Which is not correct among following?

(1) Hormones from hypothalamus reach anterior pituitary through portal circulatory system.
(2) In males, FSH and androgens regulate spermatogenesis
(3) Acromegaly, always leads to death
(4) Neurohypophysis, store and release two hormones.

51) Given below are two statements - **Statement-I :** Thyroid hormones control the metabolism of carbohydrate, protein, fat.
Statement-II : Thyroid hormones support the process of red blood cell formation.
In the above statement choose most appropriate answer from the option :-

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

52) Choose the incorrect statement about thymus gland :

(1) Present behind the sternum, ventral to dorsal aorta.
(2) Plays very important role in differentiation of T-lymphocytes and B-lymphocyte.
(3) Provides cell mediate immunity as well as humoral immunity
(4) Secretes peptide hormone thymosin

53) **Statement-I :** One pair pectoral and one pair pelvic girdle present in human body.
Statement-II : Each half of pelvic girdle is consists of two coxal bones.

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(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

54) Match the columns and choose the correct option.

	Column-I		**Column-II**
A.	Vertebral column	(i)	Flate bone
B.	Scapula	(ii)	Sesamoid bone
C.	Patella	(iii)	Sutures
D.	Cranial bones	(iv)	26 bones

	A	**B**	**C**	**D**
(1)	(iv)	(i)	(ii)	(iii)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(ii)	(iv)	(iii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

55) **Assertion (A) :** 8th, 9th, 10th pair of Ribs are vertebrochondral.
Reason (R) : These ribs do not articulate directly with sternum but joint the 7th rib with help of hyaline cartilage.

(1) Both A & R are true & R is correct explanation of A

(2) Both A & R are true but R is not correct explanation of A

(3) A is true but R is false

(4) Both A & R are false

56) Choose the incorrect statement.

(1) Mammary glands of female undergo differentiation during pregnancy.

(2) Mammary alveoli synthesises milk with help of prolactin.

(3) Estrogen stimulate formation of milk alveoli.

(4) Oxytocin causes contraction of milk alveoli to eject out the milk.

57) Given below are two statements :-
Statement-I :- Sex of the baby is determined by father and not by the mother.
Statement-II :- The presence of X or Y chromosome in the sperm determines the sex of embryo.

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Choose the correct answer from the option given below :-

(1) Both statement I and II are correct
(2) Both statement I and II are incorrect
(3) Statement I is correct and statement II is incorrect
(4) Statement I is incorrect and statement II is correct

58) Choose the incorrect statement :

(1) hCG secreted only from placenta.
(2) Parturition is regulated by neuro-endocrine mechanism.
(3) The mammary gland start producing milk towards end of pregnancy.
(4) During pregnancy concentration of estrogen, progesteron, oxytocin, prolactin, thyroxin and cortisol hormone increased several folds.

59) **Statement-I :** Primary spermatocyte is a haploid structure.
Statement-II : I^{st} polar body is a diploid structure.

(1) Both **Statement I** and **Statement II** are true.
(2) Both **Statement I** and **Statement II** are false.
(3) **Statement I** is true but **Statement II** is false.
(4) **Statement I** is false but **Statement II** is true.

60) Given below are two statements :-
Statement-I : Early symptoms of most of the sexually transmitted diseases are minor and including itching, fluid discharge, slight pain and swelling etc. in the genital region.
Statement-II : Infections or diseases which are transmitted through sexual intercourse are collectively called sexually transmitted infections or venereal diseases.
In the light of the above statements select the correct answer from the options given below-

(1) Statement-I and II are correct.
(2) Statement-I is incorrect but II is correct.
(3) Statement-I is correct but II is incorrect.
(4) Statement-I and II both are incorrect.

61) In the given list how many could be the reasons for infertility ?
Physical, congenital, diseases, drugs, immunological, Psychological.

(1) 4
(2) 6
(3) 5
(4) 3

62) Given below two statement one is labelled assertion and Reason.
Assertion (A) : The Autonomic neural system transmits impulses from the CNS to the involuntary organs and smooth muscles of the body.

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Reason (R) : The PNS comprises of all the nerves of the body associated with the CNS.

(1) Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Assertion and Reason are true and Reason is not a correct explanation of Assertion.
(3) Assertion true and Reason false.
(4) Assertion false and Reason true.

63) **Assertion (A) :** New potential developed at post synaptic membrane may be either excitatory or inhibitory.
Reason (R) : New potential developed at post synaptic membrane depends on types of neurotransmitter released by synaptic vesicles.

(1) Both (A) and (R) are correct and (R) is not the correct explanation of (A).
(2) (A) is not correct but (R) is correct.
(3) (A) is correct but (R) is not correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

64) Given below are two statements :
Statement-I : Small amounts of androgenic steroids are secreted by the adrenal cortex which play a role in the growth of axial hair, pubic hair and facial hair during puberty.
Statement-II : Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions and helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

65) **Statement-I :** Hormones which interact with membrane bound receptors normally enter the target cell and generate second messenger which in turn regulate cellular metabolism.
Statement-II : Hormones which interact with intracellular receptors mostly regulate gene expression or chromosome function by interaction of hormone receptor complex with genome.

(1) Statement-I is false & Statement-II is true.
(2) Statement-I is true & Statement-II is false.
(3) Both Statements-I & Statement-II are true.
(4) Both Statements-I & Statement-II are false.

66) Find out the correct option in given statements -
(a) Muscle is mesodermal origin.
(b) Movement of limbs, jaws and tongue require no muscular movement.
(c) Visceral muscles are located in inner wall of hollow visceral organ of body.
(d) The light band contains actin and is called I-band.

(1) a, b, c and d
(2) a, c and d

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(3) a, b, d and c

(4) a, c and b

67) Read the column-I, II and III and identify their correct match

Column-I		Column-II		Column-III	
(A)	Increased level of GnRH	(P)	Numerous mitochondria	(i)	Ovulation
(B)	Middle piece of sperm	(Q)	A tiny finger like structure	(ii)	Secretion of LH & FSH
(C)	Clitoris	(R)	Release of secondary oocyte	(iii)	Energy for tail movement
(D)	Rupturing of graffian follicle	(S)	Act on anterior pituitary	(iv)	lies at the upper junction of two labia minora

(1) (A)-P-i ; (B)-Q-ii ; (C)-R-iii ; (D)-S-iv

(2) (A)-S-ii ; (B)-P-iii ; (C)-Q-iv ; (D)-R-i

(3) (A)-R-iii ; (B)-Q-iv ; (C)-S-i ; (D)-P-ii

(4) (A)-Q-iv ; (B)-R-i ; (C)-P-ii ; (D)-S-iii

68) Match the Column I with Column II & Column III and choose the correct option

Column I		Column II		Column III	
(A)	Menstrual Phase	(i)	Rupture of follicle	(a)	Increase in FSH and LH
(B)	Follicular Phase	(ii)	Breakdown of endometrium	(b)	Secretion of progesterone
(C)	Ovulatory Phase	(iii)	Formation of corpus luteum	(c)	LH surge
(D)	Luteal Phase	(iv)	Development of follicles	(d)	Low progesterone

(1) A-iii-d, B-iv-a, C-i-c, D-ii-b

(2) A-ii-a, B-iv-d, C-i-c, D-iii-b

(3) A-ii-d, B-iv-a, C-i-b, D-iii-c

(4) A-ii-d, B-iv-a, C-i-c, D-iii-b

69) Match List-I with List-II

	List-I		List-II
(A)	In vitro fertilization	(I)	Gamete transfer
(B)	In vivo fertilization	(II)	Fertilization within the body
(C)	ZIFT	(III)	Semen into uterus
(D)	GIFT	(IV)	Embryo upto 8 blastomere
(E)	IUT	(V)	Fertilization outside the body
(F)	IUI	(VI)	Embryo more than 8 blastomere

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Choose the correct answer from the option given below:

(1) A-V, B-II, C-IV, D-I, E-VI, F-III

(2) A-V, B-I, C-II, D-III, E-IV, F-IV

(3) A-V, B-I, C-II, D-III, E-VI, F-IV

(4) A-II, B-V, C-III, D-VI, E-I, F-IV

70) Match the list-I with list-II :-

	List-I		**List-II**
A.	Site of information processing and control	I.	Somatic neural system
B.	Relays impulses from the CNS to skeletal muscles	II.	Autonomic neural system
C.	Comprises of all the nerves of the body associated with the CNS	III.	Central neural system
D.	Transmits impulses from the CNS to the involuntary organs	IV.	Peripheral neural system

Choose the correct answer from the option given below :-

(1) A-(III), B-(I), C-(IV), D-(II)

(2) A-(III), B-(II), C-(IV), D-(I)

(3) A-(IV), B-(I), C-(II), D-(III)

(4) A-(I), B-(II), C-(III), D-(IV)

71) **Assertion :** The cerebral cortex is reffered as grey matter due to its greyish appearance.
Reason : The neuron cell bodies are concentrated here giving the colour.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

72) Column-I list the parts of the human brain and column-II lists the functions. Match the two columns and identify the correct choice from those given :

	Column I		**Column II**
A	Cerebrum	p	controls the pituitary
B	Cerebellum	q	controls vision & hearing reflex
C	Hypothalamus	r	controls rate of heartbeat
D	Midbrain	s	Seat of intelligence
		t	maintains body posture

(1) A=t, B=s, C=q, D=p

(2) A=s, B=t, C=q, D=p

(3) A=t, B=s, C=p, D=q

(4) A=s, B=t, C=p, D=q

73) How many glands which are given below considered in endocrine system.
Pituitary, pineal, thyroid, Adrenal, Pancreas, Parathyroid, Thymus, Gonads.

(1) Three

(2) Eight

(3) Seven

(4) Five

74) Which of the following is correct?
(A) Thyrocalcitonin (TCT) is hypercalcemic hormone.
(B) Parathyroid hormone (PTH) increases the Ca^{+2} levels in the blood.
(C) Thymosins play a major role in the differentiation of T-lymphocytes.
(D) Parathyroid glands are present on back side of the thyroid gland.

(1) B, C, D

(2) A, B, C

(3) A, C, D

(4) A, B, C, D

75) Calcium balance in the body is regulated by __A__ and __B__.

	A	B
1.	T_3	PTH
2.	PTH	T_4
3.	PTH	TCT
4.	TCT	T_4

(1) 1

(2) 2

(3) 3

(4) 4

76) Thymosins play a major role in the differentiation of –

(1) T–lymphocytes which provide cell-mediated immunity.

(2) T-lymphocytes which provide innate immunity.

(3) B-lymphocytes which provide cell-mediated immunity.

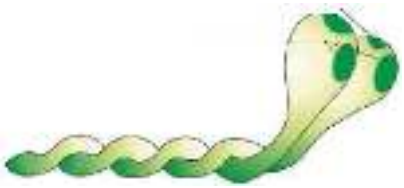
(4) Corpus luteum which secretes progesterone.

77) Identify the wrong statements:

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(A) The atrial wall of our heart secretes atrial natriuretic factor which increase blood pressure.
(B) Estrogen is secreted mainly by the growing ovarian follicles.
(C) Insulin is a steroid hormone, which plays a major role in the regulation of glucose homeostasis.
(D) Prolonged hyperglycaemia leads to a complex disorder called diabetes mellitus.
Choose the most appropriate answer from the options given below:

(1) D and E only
(2) A and B only
(3) C and E only
(4) A and C only

78) Study the given diagram and select the correct option for it. 

(1) It is a meromyosin which form the thin filament
(2) It is a meromyosin which is monomer of thick filament
(3) It has two parts a globular head with cross arm and a heavy meromyosin
(4) It form myosin filament which found in I-band of muscle

79) Which of following option is correct for locomotion.

(1) Vary with their habitats and the demand of the situation
(2) It is generally for search of food and shelter
(3) Structures which are involved in locomotion, need not be different from those affecting other types of movements
(4) All of these

80) **Assertion (A) :** Adequate calcium essential for maintaining muscle contraction.
Reason (R) : Calcium is necessary for removal of troponin - tropomyosin complex from actin.

(1) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(2) Both Assertion and Reason are true, but Reason is NOT the correct explanation for Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

81) Select the incorrect match.

(1)	Ball and socket joint	between humerus and pectoral girdle
(2)	Pivot joint	between atlas and axis
(3)	Gliding joint	between the carpals
(4)	Hinge joint	between carpal and metacarpal of thumb

(1) 1
(2) 2

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(3) 3

(4) 4

82) Choose the option which is not correct w.r.t. the disorders of muscular and skeletal system.

(1) Arthritis is the inflammation of joints

(2) Gout is the inflammation of joints due to accumulation of lactic acid crystals

(3) Muscular dystrophy is the progressive degeneration of skeletal muscle mostly due to genetic disorder

(4) Osteoporosis is an age-related disorder characterised by decreased bone mass and increased chances of fractures

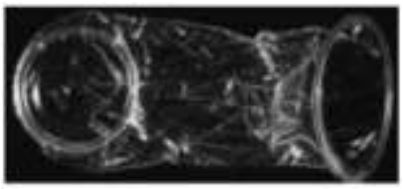
83) **Statement 1 :** Joints are essential for all types of movements involving the bony part of the body.
Statement 2 : Joint are points of contact between bones, or between bones and cartilage.

(1) Both statement are incorrect

(2) Both statement are correct

(3) Only statement 1 is correct

(4) Only statement 2 is correct

84) Select the correct statement for the given diagram –  a. It is used as contraceptive
b. It is disposable
c. It protects the user from contracting STIs
d. It is made of thin rubber/latex sheath that are used to cover the vagina and cervix in the female

(1) Only a, b, c

(2) Only a, b, d

(3) a, b, c, d

(4) Only a, c, d

85) Which technologies in the list given below are showing in-vivo fertilisation?
GIFT, ZIFT, ICSI, IUT, AI, IUI, IVF

(1) GIFT, ZIFT, ICSI, IUT

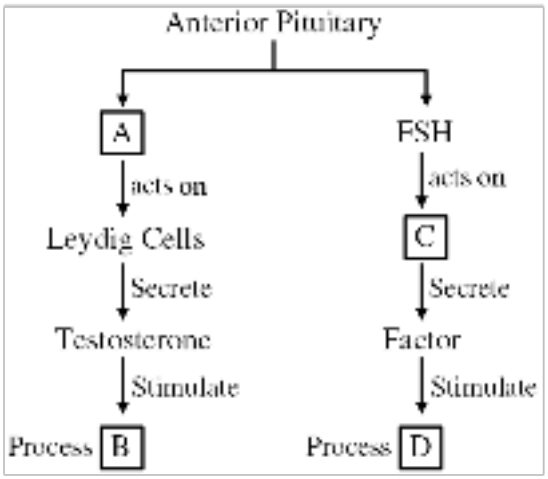
(2) GIFT, AI, IUI

(3) IUT, AI, IUI, IVF

(4) ZIFT, ICSI, IUT

86) Following is the flow chart showing the influence of hormones on testes in males. Choose the option which correctly fills the gaps represented by A, B, C and D.

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(1) A-Andogen; B-Spermiation;
C-Seminiferous tubule; D-Spermatogensis

(2) A-Inhibin; B-Spermiogenesis;
C-Spermatogonia; D-Spermatogenesis

(3) A-ICSH/LH; B-Spermatogenesis;
C-Sertoli cells; D-Spermiogenesis

(4) A-LH; B-Spermiogenesis;
C-Sertoli cells; D-Spermatogenesis

87) In a menstrual cycle, hormones start increasing in which order from the beginning of menstruation ?

(1) FSH, progesterone, estrogen

(2) Estrogen, FSH, progesterone

(3) FSH, estrogen, progesterone

(4) Esterogen, progesterone, FSH

88) Which of the following statement is not incorrect.

(1) Only endometrium undegoes cyclic changes during the menstrual cycle.

(2) Only Cervical canal is called birth canal.

(3) Opening of vagina is often partially covered by membrane called hymen.

(4) Clitoris is a tiny finger-like structure which lies at upper junctions of labia majora

89) Which of the following hormone's level is increased during pregnancy in the maternal blood ?
(a) FSH
(b) Progesterone
(c) Thyroxine
(d) Cortisol
(e) LH
(f) Estrogen

(1) a, b, e, f

(2) a, b, c, d, e

(3) c, d, a

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(4) b, d, c, f

90) Which of the following is incorrectly paired with its function.

(1)	Ovary	Synthesis and secretion of steroid(estrogen) hormone
(2)	Fimbriae	Collection of the ovum after ovulation
(3)	Seminal vesicle	Produces a sugar containing fluid to nourish sperm
(4)	Bartholin glands	Secrete alkaline fluid to destroys the acidity of the male urethrra.

(1) 1

(2) 2

(3) 3

(4) 4

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	3	3	1	3	4	1	3	2	3	1	3	1	4	1	1	2	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	3	1	2	3	1	3	2	1	2	2	1	3	4	1	3	4	3
Q.	41	42	43	44	45															
A.	2	2	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	2	1	4	3	2	4	2	1	2	4	2	2	3	4	4	2	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	1	2	4	3	3	3	4	4	3	1	2	3	3	2	3	1	2	2
Q.	86	87	88	89	90															
A.	3	4	4	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	4	2	2	1	4	1	3	3	1	1	3	2	3	3	2	3	3	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	2	4	2	2	1	1	3	3	1	3	2	4	3	4	2	3	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	4	3	2	2	3	4	3	3	1	2	2	1	1	3	1	4	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	1	4	1	2	2	4	1	1	1	4	2	1	3	1	4	2	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	2	3	2	3	3	3	4	4										

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PHYSICS

1)

Assertion: True — speed decreases as particle moves away from mean. which varies with time because .T(t) is time-

• Reason: True — acceleration in SHM is a — ω^2x (t) dependent.

2) If m is the mass and r is the amplitude of oscillation, then maximum kinetic energy,

$$k_0 = \frac{1}{2}m\omega^2 r^2$$

$$r = \left(\frac{2k_0}{m\omega^2}\right)^{\frac{1}{2}}$$

The displacement equation can be

$$y = r\sin\omega t = \left(\frac{2k_0}{m\omega^2}\right)^{\frac{1}{2}}\sin\omega t$$

3) $F = -\frac{dU}{dx}$

F = –4bx

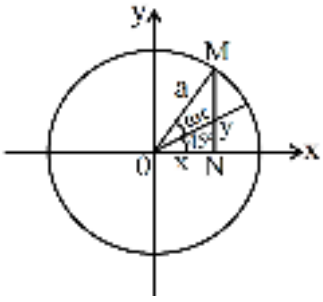
–kx = –4bx

k = 4b

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$T = 2\pi\sqrt{\frac{m}{4b}}$$

$$T = \pi\sqrt{\frac{m}{b}}$$



4)

In ΔOMN

$$\cos(\omega t + 45^o) = \frac{x}{a}$$

$$x = a\cos(\omega t + 45^o)$$

$$x = a\cos\left(\frac{2\pi t}{4} + \frac{\pi}{4}\right)$$

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5) $T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow m \propto T^2$

$\Rightarrow \frac{m_2}{m_1} = \frac{T_2^2}{T_1^2} \Rightarrow \frac{M+m}{M} = \left(\frac{\frac{5}{4}T}{T}\right)^2$

$\Rightarrow \frac{m}{M} = \frac{9}{16}$

6) Let n be the frequency of fork C then

$n_A = n + \frac{3n}{100} = \frac{103n}{100}$ and $n_B = n - \frac{2n}{100} = \frac{98n}{100}$

but $n_A - n_B = 5 \Rightarrow \frac{5n}{100} = 5 \Rightarrow n = 100$ Hz

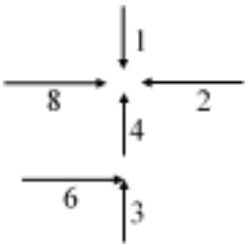
$\therefore n_A = \frac{(103)(100)}{100} = 103\text{Hz}$

7)

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

for $I_{max} \Rightarrow \phi = 0 \Rightarrow I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

8)



$A_{net} = \sqrt{6^2 + 3^2} = \sqrt{45}$ $\quad \tan\theta = \frac{1}{2}$

9) $v = \sqrt{gx}$

$\frac{dx}{dt} = \sqrt{g}(x)^{1/2}$

$\int_0^{\ell} x^{\frac{-1}{2}} dx = \sqrt{g}\int_0^{\ell} dt$

$2\sqrt{\ell} = \sqrt{g}\,t$

$t = 2\sqrt{\frac{\ell}{g}}$

10)

Let the two consecutive resonance frequencies correspond to harmonics n and n + 1

$f_n = \frac{nV}{2L} = 375, \quad f_{n+1} = \frac{(n+1)V}{2L} = 450$

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Subtracting: $\frac{V}{2L}$ = 450 – 375 = 75 Hz (fundamental frequency)

Wave speed $V = \sqrt{\frac{360}{4 \times 10^{-3}}} = \sqrt{90000} = 300\,m/s$

Using $\frac{V}{2L}$ = 75: 2L = $\frac{300}{75}$ = 4m ⇒ L = 2m

Mass m = μL = $(4 \times 10^{-3}) \times 2 = 8 \times 10^{-3}$ kg = 8 g.

11) 1st 2nd 3rd ..... n^{th}

256 256+(2–1)×4 256+(3–1)×4 256+(n–1)×4

256 + (n–1) × 4 = 2 × 256

⇒ (n–1) × 4 = 256

⇒ n = 65

12) $\frac{\lambda}{4} = \ell_1 + e$ & $\frac{3\lambda}{4} = \ell_2 + e$

⇒ $\lambda = 2(\ell_2 - \ell_1)$

speed of sound = fλ

v = 512 × 2 × 34 × 10^{-2} = 348 m/s

13) $Q_{ethyl} = (Q)_{water}$

$m\frac{s}{2}\Delta T = m \times s \times 25$

$\frac{\Delta T}{2} = 25$

ΔT = 50°C

14) PV = nRT

$\frac{V}{T} = \frac{nR}{P}$

Slope = $\frac{nR}{P_1} < \frac{nR}{P_2}$

$P_2 < P_1$

15)

$P^{\gamma-1} \propto T^{\gamma}$

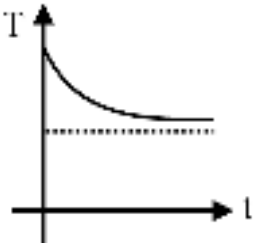
$P \propto T^{\left(\frac{\gamma}{\gamma-1}\right)}$

On Comparing

$C = \frac{\gamma}{\gamma - 1} = \frac{7/5}{7/5 - 1} = \frac{7}{2}$

16)

A/C to Newton's law of cooling



17) dQ = 800 kJ ; ΔT = 120 – 20 = 100°C
m = 8 kg ; S = ?
$S = \frac{dQ}{m\Delta T} = \frac{800}{8 \times 100}$ = 1 kJ/kg °C

18) α = linear expansion
γ = 3α
$\alpha = \frac{\gamma}{3}$; $\gamma^1 = 3 \times \frac{\gamma}{3}$
$\gamma^1 = \gamma$
Volume expansion remain unchanged.

19)

Eq. of st. line y = mx + C
$P = -\frac{P_0}{V_0}V + 3P_0$
$\frac{nRT}{V} = -\frac{P_0}{V_0}V + 3P_0$
$T = \frac{-P_0}{V_0 nR}V^2 + \frac{3P_0}{nR}V$ .....(i)
for maximum temperature $\frac{dT}{dV} = 0$
$\frac{dT}{dV} = \frac{-P_0}{V_0(nR)}(2V) + \frac{3P_0}{nR}(1) = 0$
$V = \frac{3}{2}V_0$ (put value in equation (1))
So, $T = \frac{-P_0}{V_0 nR}\left(\frac{9}{4}V_0^2\right) + \frac{3P_0}{nR}\left(\frac{3V_0}{2}\right)$
$T = -\frac{9}{4}\frac{P_0V_0}{nR} + \frac{9}{2}\frac{P_0V_0}{nR}$
$T = \frac{9P_0V_0}{4nR}$

20) $\left(\frac{dQ}{dt}\right)_A = \left(\frac{dQ}{dt}\right)_B$
$\frac{300A(100-T)}{\ell} = \frac{200A(T-0)}{\ell}$
300 – 3T = 2T
5T = 300

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T = 60°C

21) $m \times L_f + ms\Delta T + Ms\Delta T = 6 \times 10^3$
$50 \times 80 + 50 \times 1 \times (T - 0) + 200 (T - 0) = 6 \times 10^3$
$4000 + 250\ T = 6000$
$250T = 2000$
T = 8°C

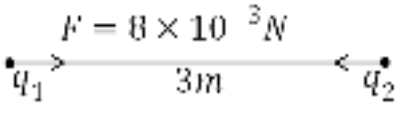
22) According to Stefan-Boltzmann law, the energy radiated per second through the surface of area A is given by
$E = \sigma AT^4$
$\therefore \frac{E_1}{E_2} = \frac{A_1}{A_2}\left(\frac{T_1}{T_2}\right)^4$
or $10000 = \frac{r_1^2}{r_2^2}\left(\frac{2000}{6000}\right)^4$
or , $\frac{r_1^2}{r_2^2} = (30)^4$
or $r_1 : r_2 = 900 : 1$

23)

Flux through surface S_1 :
$(\phi_1) = \frac{\text{Charge enclosed within } S_1}{\varepsilon_0} = \frac{q}{\varepsilon_0}$
Since the shell is conducting therefore a charge of magnitude -q will be induced at the inner surface (radius R_1) of the shell.
$\therefore$ Flux through surface S_2
$(\phi_2) = \frac{\text{Charge enclosed within } S_2}{\varepsilon_0} = \frac{q-q}{\varepsilon_0} = 0$

24) $K = \frac{\sigma}{\sigma - \rho} = \frac{1.6}{1.6 - 0.8} = 2$
$\sigma \rightarrow$ density of material of sphere
$\rho \rightarrow$ density of liquid

25) $\frac{ke^2}{r} = \frac{1}{2}mv^2 \times 2$
$v = \sqrt{\frac{ke^2}{mr}}$

26) 

We have to calculate charge 'q' and 'q_2'.

$q_1 + q_2 = 6 \times 10^{-6}$ C

$\frac{Kq_1q_2}{3^2} = 8 \times 10^{-3}$ N

$\frac{9 \times 10^9 \times q_1q_2}{3^2} = 8 \times 10^{-3}$

$q_1q_2 = 8 \times 10^{-12}$

$(q_1 - q_2)^2 = (q_1 + q_2)^2 - 4q_1q_2$

$(q_1 - q_2)^2 = 36 \times 10^{-12} - 4 \times 8 \times 10^{-12}$

$q_1 - q_2 = 2 \times 10^{-6} = 2\mu C$ .....(i)

$q_1 + q_2 = 6 \times 10^{-6} = 6\mu C$.....(ii)

From (i) & (ii), we get

$q_1 = 4\mu C, q_2 = 2\mu C$

Final Answer: Option (2)

27)

$dq = \rho.dv = \rho.4\pi r^2.dr$

$\Rightarrow dq = \frac{Q.r}{\pi R^4}.4\pi r^2.dr$

$= \frac{4Qr^3}{R^4}.dr$

Therefore,

$q = \int_0^{r_1} \frac{4Qr^3}{R^4}.dr$

$\Rightarrow q = \frac{Qr_1^4}{R^4}$

Now, using gauss law,

$E(4\pi r_1^2) = \frac{q}{\in_0}$

$\Rightarrow E = \frac{q}{4\pi \in_0 r_1^2}$

$\Rightarrow E = \frac{Qr_1^2}{4\pi \in_0 R^4}$

28)

Because, $\sigma = \frac{Q}{4\pi R^2} \Rightarrow \frac{q_1}{q_2} = \left(\frac{R_1}{R_2}\right)^2 = \frac{1}{4}$

and $q_1 + q_2 = Q \Rightarrow \frac{q_2}{4} + q_2 = Q$

$q_2 = \frac{4Q}{5}$

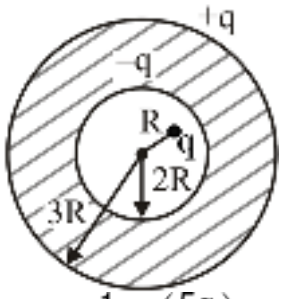
29) At centre

$v_0 = v_1 + v_2 + v_3$

$= \frac{kq}{R} + \frac{k(-q)}{2R} + \frac{kq}{3R}$

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$$v_0 = \frac{1}{4\pi\varepsilon_0}\left(\frac{5q}{6R}\right)$$

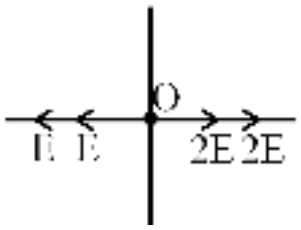
30)

We know that field due to dipole at equatorial

$$E_{eq} = \frac{KP}{r^3} = E \text{ (direction } (+) \rightarrow (-))$$

at axis,

$$E_{axis} = \frac{2KP}{r^3} = 2E(\text{direction } (-) \rightarrow (+))$$



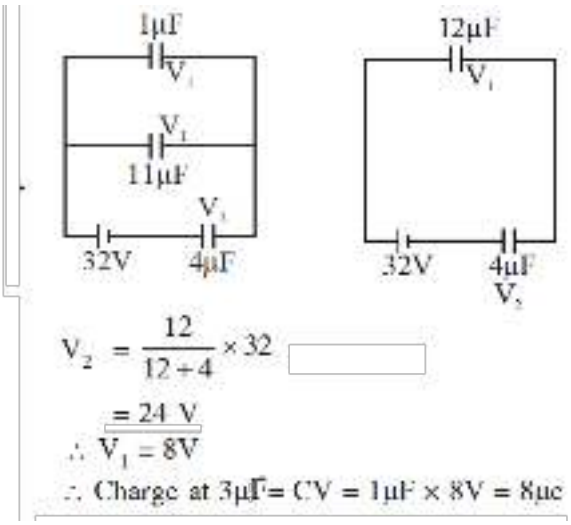
So, $E_{net} = 4E - 2E = 2E = \frac{2KP}{r^3}$

31) $E = \frac{V}{d}$

$E \propto \frac{1}{d}$, V→same, d↑, E↓

C↓, q = CV ↓

32)



$$V_2 = \frac{12}{12+4} \times 32$$

$$= 24 \text{ V}$$

$$\therefore V_1 = 8V$$

∴ Charge at 3μF = CV = 1μF × 8V = 8μc

33) (n–1) capacitors are in series combination.

$$C_1 = \frac{C}{n-1}$$

$$C_{eq} = \frac{C}{n-1} + C = \frac{C + (n-1)C}{n-1} = \frac{Cn}{n-1}$$

34)

Charge on 6μF capacitor = 10μC
equivalent capacitance of 2μF, 3μF and 6μF series combination = 1μF

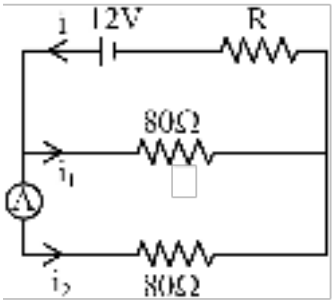
Potential difference across above series combination of three capacitors, $V = \frac{10\mu C}{1\mu F} = 10$ volts
The same P.D. is across 4μF capacitor. So, charge on it = 4μF × 10 V = 40 μC

35)

Equivalent capacitance of C_1, C_2 & C_3 in parallel, $C_P = C_1 + C_2 + C_3 = 12$ units Also, $C_1 . C_2 . C_3 = 48$ and $C_1 + C_2 = 6$ units
Therefore, $C_3 = 12 - 6 = 6$ units

Now, $C_1C_2 = \frac{48}{C_3} = \frac{48}{6} = 8$
Therefore,
$(C_1 - C_2)^2 = (C_1 + C_2)^2 - 4C_1C_2$
$= 6^2 - 4 \times 8 = 4$
$\Rightarrow C_1 - C_2 = 2$
So, $C_1 = 4$ units, $C_2 = 2$ units, $C_3 = 6$ units

36)



$i_1 = i_2 = 0.1$ A (equal resistance)
$i = i_1 + i_2 = 0.2$ A
From figure

$$i = \frac{12}{R + 40}$$

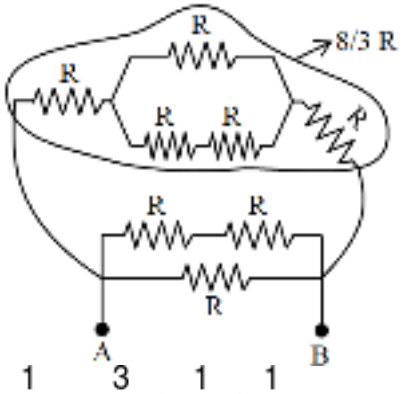
$$R + 40 = \frac{12}{0.2} = 60 \Rightarrow R = 20\ \Omega$$

37)

Use symmetry method
Effect circuit is $\Rightarrow R_1R_4 = R_2R_3$

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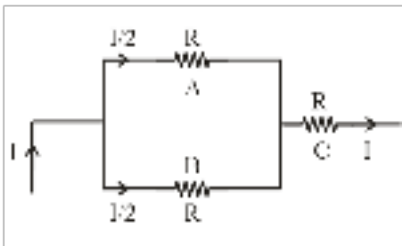


$$\frac{1}{R_{AB}} = \frac{3}{8R} + \frac{1}{2R} + \frac{1}{R}$$

$$= \frac{3+4+8}{8R} = \frac{15}{8R}$$

$$R_{AB} = \frac{8}{15}R = \frac{8}{15}(5) = \frac{8}{3}\Omega \quad (\because R = 5\ \Omega)$$

38)



$P = I^2 R$

$$P_A : P_B : P_C = \left(\frac{I}{2}\right)^2 : \left(\frac{I}{2}\right)^2 : I^2$$

$= 1 : 1 : 4$

39)

Nichrome and Manganin have Low temperature coefficient of resistance.

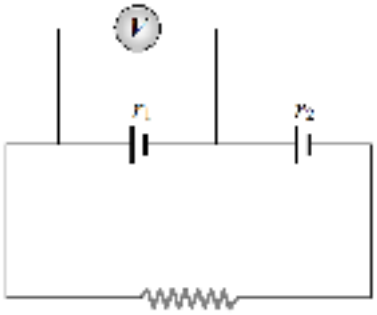
40) $$\frac{R}{S} = \frac{\ell}{100-\ell} \Rightarrow \frac{2}{3} = \frac{\ell}{100-\ell}$$

$\ell = 40$ cm

41) $$R = \frac{V}{I} = \tan\theta = \tan 30^\circ = \frac{1}{\sqrt{3}}$$ ohm

42) Let the voltage across cell one is V, then

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$$V_1 = E_1 - ir_1 = E - r_1\left(\frac{2E}{r_1 + r_2 + R}\right)$$

But $V_1 = 0$

$$\Rightarrow E - \frac{2Er_1}{r_1 + r_2 + R} = 0$$
$$\Rightarrow r_1 + r_2 + R = 2r_1$$
$$\Rightarrow R = r_1 - r_2$$

43)

$$i = \frac{E_{net}}{R_{net}} = \frac{10-4}{1+2+3} = \frac{6}{6} = 1A$$

44)

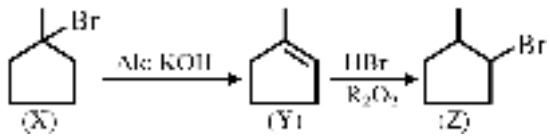
At null point deflection through galvanometer is zero, also deflection in galvanometer in opposite direction, either side of null point.

45)

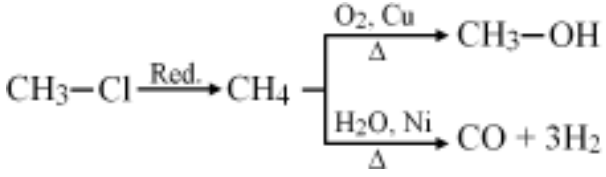
Its a balanced WSB.

CHEMISTRY

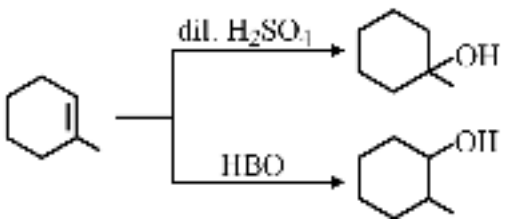
46)



47)



48)

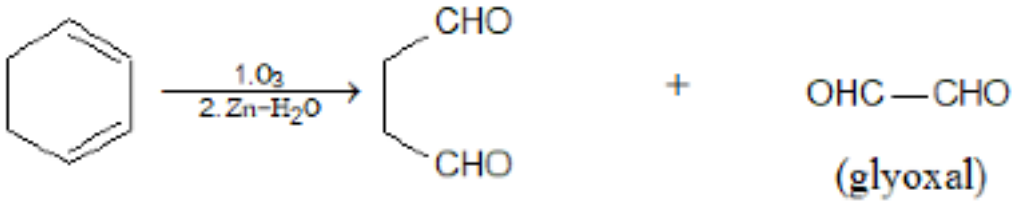


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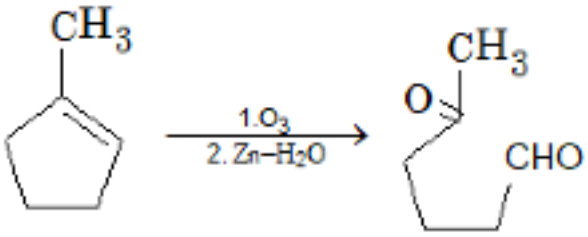
49)

Boiling pt $\alpha \frac{1}{\text{Surface Area}}$

so Ans: A



ethane-1, 2-dicarbaldehyde



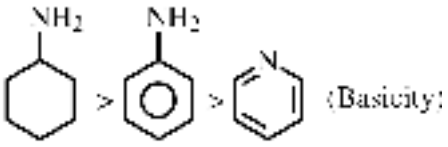
50) 5-oxohexanal

51)

Lone pair acceptor → Electrophile

52)

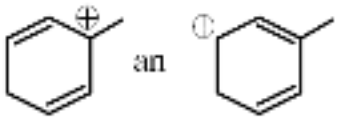
SB > Pd b > db (BL order)

53) 

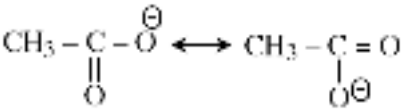
54)

HOC ∝ No. of carbon

55)

(1) 

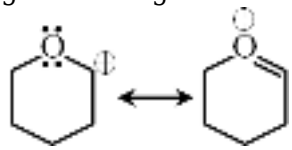
both have resonance and hyper conjugation effect but structure (I) 3° carbocation is more stabile than 2° carbocation

(2) 

both are equivalent resonating structure so equal contribution.

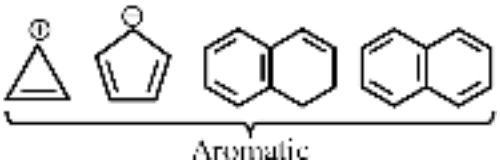
(3) 

Negative charge on more EN. atom will be more stable

(4) 

Second have more No. of covalent bond

Final Answer :- (1)

56) 

57)

Localised LP > Delocalised LP (Basicity)

58)

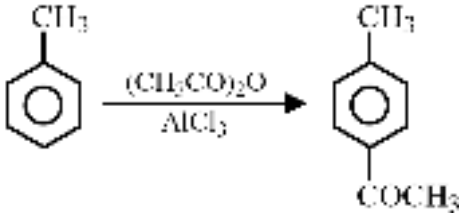
$$P + Na_2O_2 \rightarrow Na_3PO_4 + 2Na_2O$$
$$Na_3PO_4 + 3HNO_3 \rightarrow H_3PO_4 + 3NaNO_3$$
$$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow$$
$$(NH_4)_3PO_4 \cdot 12MoO_3 + 2NH_4NO_3 + 12H_2O$$

(Yellow ppt)

Ammonium phosphomolybdate

A yellow colouration or precipitate indicate the presence of phosphorus

Final Answer : Yellow ppt of $(NH_4)_3PO_4.12MoO_3$

59) 

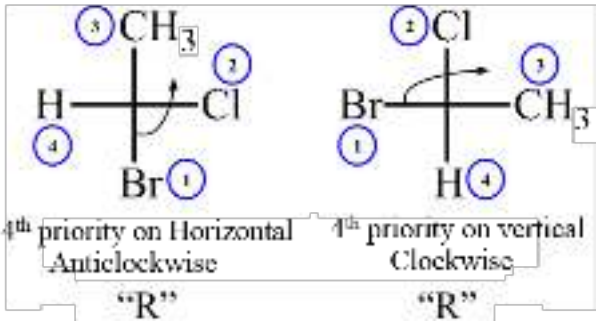
60) Knowledge based

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Distillation

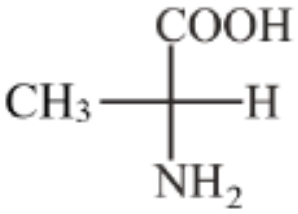
61)

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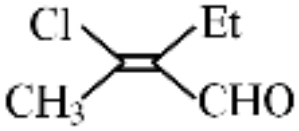


Conclusion: Both structures have the **(R)-configuration**. Identical configurations imply they represent the same molecule (Homomers).

Final Answer: Option (4) Homomers



62)
'R' configuration

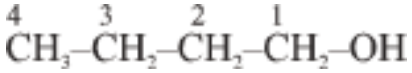


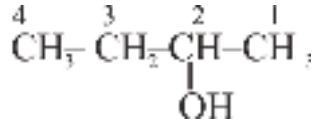
63)
'E' configuration

64)

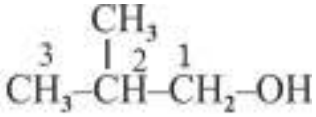
All have same 'S' configuration hence all are identical.

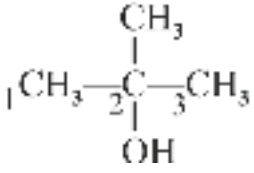
65)





and





and

Final answer : Option(1)

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66)

Position of double bond is different So position isomers

67)

Equivalents of N – (Equi. H_2SO_4 initial) – (Equiv. NaOH used)

A. W – 0.1 g
B. Initial H_2SO_2 : 0.2 N × 25 mL
C. NaOH excess 0.1N × 20 mL
D. Equivalent weight of N = 14.

A. Calculate Equivalent (Eq) of H_2SO_4 consumed by $NH_3(Eq_N)$

Eq_N = (0.2N × 25mL) - (0.1 N × 20 mL)
Eq_N = 5 mEq - 2mEq = 3mEq = 0.003 Eq

2. Calculate Percentage of Nitrogen

$$\text{Percentage of N} = \frac{Eq_N \times \text{Equivalent of N}}{W} \times 100$$

$$\text{Percentage of N} = \frac{0.003\text{Eq} \times 14\text{g/Eq}}{0.1\text{g}}$$

Percentage of N – 42 %

Final Answer: Option (3)

68) **Question Explanation :** Find the order n of the reaction using how half-life changes with initial pressure p.

Concept : This question is based on $t_{1/2} \propto (p)^{1-n}$.
Solution : Given :
$p_1 = 1\ \text{atm}, t_1 = 100\ \text{s}$
$p_2 = 0.5\ \text{atm}, t_2 = 50\ \text{s}$
Use :
$t_{1/2} \propto (p)^{1-n}$
Take ratio :

$$\frac{t_1}{t_2} = \left(\frac{p_1}{p_2}\right)^{1-n}$$

$$\frac{100}{50} = \left(\frac{1}{0.5}\right)^{1-n}$$

$$2 = 2^{1-n}$$

Thus,
1 = 1 – n
n = 0
Final Answer : 0

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69)

Question Explanation: We need the percentage of reactant remaining after 24 hours for a first-order reaction with half-life 12 hours.

Concept:
This question is based on first-order kinetics (half-life relationship).

Given Data:
$t_{1/2}$ = 12 h
t = 24 h

Solution:
Formula: $a_t = a_0/2^n$, where $n = t/t_{1/2}$

$n = 24/12 = 2$
$a_t = a_0/2^2 = a_0/4 = 0.25\ a_0$

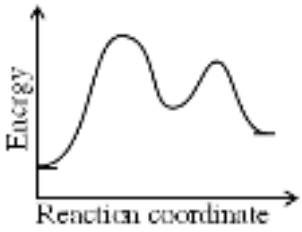
Percentage remaining = $(a_t/a_0) \times 100 = 25\%$

Final Answer: Option (2)

70) For same temperature change

$$\ell n\left(\frac{k_2}{k_1}\right) \propto E_a$$

$$\boxed{\mu \propto E_a}$$

71) 

72) $r = K[A]^x[B]^y$
$0.045 = K(0.05)^x(0.05)^y$ ....(1)
$0.090 = K(0.10)^x(0.05)^y$ ....(2)
$0.72 = K(0.20)^x(0.10)^y$ ....(3)

From (1) ¸ (2), $\frac{0.045}{0.090} = \left(\frac{0.05}{0.10}\right)^x \Rightarrow x = 1$

From (2) ÷ (3), $\frac{0.090}{0.720} = \left(\frac{0.10}{0.20}\right)^x \cdot \left(\frac{0.05}{0.10}\right)^y \Rightarrow y = 2$

Hence, $r = K[A][B]^2$
Correct option : (3)

73) **Question Explanation :** We are asked to identify the correct graph of rate of consumption (R) vs time for a first-order reaction.
Concept : First-order reaction kinetics

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Solution : For a first-order reaction: $R = -d[A]/dt = k[A]$
As time progresses, [A] decreases exponentially, so the rate also decreases exponentially.
Thus, the graph of R vs t is an exponentially decreasing curve.
Final Answer : Option 3

74) $v.p \propto \frac{1}{C_{net}}$

75) $\Delta T_f = K_f m$ ...(1) $\Delta T_f = T_f° - T_f$
$\Delta T_b = K_b m$ ...(2)
(1) ÷ (2)
$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b} \Rightarrow \Delta T_b = ?$

76) Given w = 0.2g, W = 20 g, $\Delta T = 0.45$;
$\Delta T_f = \frac{1000 \times K_f \times w}{M \times W}$
or $0.45 = \frac{1000 \times 5.12 \times 0.2}{20 \times M}$
 M(observed) = 113.78
Now for $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$
Before association 1 0
After association 1–α α/2 ;
where α is the degree of association.
$\frac{M_{normal}}{M_{observed}} = 1 - \alpha + \frac{\alpha}{2}$;
or $\frac{60}{113.78} = 1 - \alpha + \frac{\alpha}{2}$; α = 0.945

77) $y_A = \frac{P_A^o x_A}{P}$ and $y_B = \frac{P_B^o x_B}{P}$
$\frac{y_A}{y_B} = \frac{P_A^o}{P_B^o} \times \frac{x_A}{x_B} \Rightarrow \frac{4}{3} = \frac{1}{3} \times \frac{1-x_B}{x_B}$
$x_B = \frac{1}{5}$

78) **Explain Question :**
Calculate freezing point of solution
Concept :
Depression in freezing point (ΔT_f) with 'i'.
Solution :
$HX \rightleftharpoons H^+ + X^- \ (n = 2)$
$i = 1 + \alpha(n-1)$
= 1 + 0.2 (2 –1)
i = 1.2
$\Delta T_f = ik_f m$

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$T_f^0 - T_f = 1.2 \times 1.86 \times 0.2$
$273 - T_f = 0.4464$
$T_f = 273 - 0.4464$
$= 272.5536K$
Final Answer option (2)

79)

in air, $P_{N_2} = 0.8\,atm,\ P_{O_2} = 0.2$ atm
$P = K_H X$ (Henry's law)
$0.8 = 2 \times 10^4\ X_{N_2}$ .....(1)
$0.2 = 10^4\ X_{O_2}$ .....(2)
Ratio of solubility of N_2 & O_2 is 2 : 1

80)

Question Explanation :
Find the total vapour pressure denoted by the point 'X' from graph.
Concept :
This question is based on Raoult's Law.
Solution :
$P^o_A = 50mm$, $P^o_B = 30mm$
$X_A = 0.2$; $X_B = 0.8$
$P_T = P^o_A X_A + P^o_B X_B = 50 \times 0.2 + 30 \times 0.8 = 34mm$
Final answer : (3)

81) **Question Explanation:** The question asks to determine the EMF (Electromotive Force) for a given concentration cell involving hydrogen gas at two different pressures, P_1 and P_2.
Concept : Nernst equation : $E_{cell} = E^0_{cell} - \frac{RT}{nF} \ln Q$
Solution :
• Anode (Oxidation): $H_2(g, P_1) \rightarrow 2H^+(aq) + 2e^-$
• Cathode (Reduction): $2H^+(aq) + 2e^- \rightarrow H_2(g, P_2)$
The overall cell reaction is: $H_2(g, P_1) \rightarrow H_2(g, P_2)$
For this reaction :
• n = 2
• $E^0_{cell} = 0$.
• $Q = \frac{P_2}{P_1}$
Substituting these values into the Nernst equation:
$E_{cell} = 0 - \frac{RT}{2F} \ln\left(\frac{P_2}{P_1}\right)$
$E_{cell} = \frac{RT}{2F} \ln\left(\frac{P_1}{P_2}\right)$
Final answer : Option 2

82) **Explain Question:**

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We have to calculate the total charge (in coulombs) required to reduce 1 mole of MnO_4^- to Mn^{2+}.

Concept:

Charge required = (number of electrons transferred) × (Faraday constant).

Solution:

Reduction of permanganate ion in acidic medium is:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

So, 1 mole of MnO_4^- requires 5 moles of electrons.

Charge required

$= 5 \times 96500$

$= 4.83 \times 10^5$ C

Final Answer: Option (3)

83) (i) $Cu + 2H^+ \rightarrow Cu^{2+} + H_2$

$$E^o_{cell} = E^o_{H^+/H_2} - E^o_{Cu^{2+}/Cu}$$

$$= 0 - 0.34$$

$$E^o_{cell} = -ve, \quad \Delta G^o = +ve$$

(ii) $Zn + 2H^+ \rightarrow Zn^{2+}, H_2$

$$E^o_{cell} = +0.76\,V$$

(iii) $2Ag + 2H^+ \rightarrow 2Ag^+ + H_2$ $\qquad E^o_{cell} = -0.80\,V$

For spontaneous reaction, emf of cell must be positive.

84) **Question Explanation :** A 5 A current is passed through a zinc sulphate solution for 40 minutes. The question asks to calculate the mass of zinc deposited at the cathode.

Concept : This question is based on Faraday's First Law of Electrolysis

Solution : Given: Current (I) = 5A; Time (t) = 40 minutes = 40 x 60 s = 2400 s

$$w = \frac{Elt}{96500} = \frac{\left(\frac{65.38}{2}\right) \times 5 \times 40 \times 60}{96500} = 4.065\ gm$$

Final answer : 2

85)

Question Explanation: By **Faraday's Second Law,** same electricity passed → same number of Faradays used.

Concept: Weight liberated ∝ equivalent weight, Equivalent weight = Molar mass / n-factor.

Solution:

1. For hydrogen gas (H_2): $2H^+ + 2e^- \rightarrow H_2$

Equivalent weight of H_2 = 2 / 2 = **1 g/eq**

2. For magnesium (Mg): $Mg^{2+} + 2e^- \rightarrow Mg$

Equivalent weight of Mg = 24 / 2 = **12 g/eq**

Ratio of weights:

Weight(H_2) / Weight(Mg) = E(H_2) / E(Mg) = 1 / 12

Final Answer : Option (2)

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86) At equilibrium, E = 0, so the Nernst equation becomes:

$$0 = E^o - \frac{0.0592}{n}\log K$$

The number of electrons transferred, n, is 2.

Rearrange the equation to solve for K:

$$\log K = \frac{nE^o}{0.0592}$$

$$\log K = \frac{2 \times 0.037}{0.0592}$$

$\log K = 1.25$

$K = 10^{1.25} = 17.78$

Initial: $[Cd^{2+}] = 1.0$ M, $[Fe^{2+}] = 0$ M

Change: $[Cd^{2+}] = -x$, $[Fe^{2+}] = +x$

Equilibrium: $[Cd^{2+}] = 1.0 - x$, $[Fe^{2+}] = x$

The equilibrium constant expression is:

$$K = \frac{[Fe^{2+}]}{[Cd^{2+}]}$$

$$17.78 = \frac{x}{1.0 - x}$$

Solve for x:

$17.78(1.0 - x) = x$

$17.78 - 17.78x = x$

$17.78 = 18.78x$

$$x = \frac{17.78}{18.78} = 0.947$$

$[Cd^{2+}] = 1.0 - x$

$[Cd^{2+}] = 1.0 - 0.947$

$[Cd^{2+}] = 0.053$ M

87) **Explain the Question :** Find moles of electrons (Faradays) needed to oxidize 1 mol $C_5H_{12} \rightarrow CO_2$.

Concept : This question is based on compute oxidation state change per C, multiply by number of C atoms.

Solution :

Let oxidation state of C in C_5H_{12} be x.

$5x + 12(+1) = 0 \Rightarrow 5x = -12 \Rightarrow x = -\frac{12}{5} = -2.4$

In CO_2, C = +4. Change per C = +4 – (–2.4) = 6.4 electrons.

Total electrons for 5C atoms = 6.4 × 5 = 32 mole e^-.

So charge = 32 Faradays

Final Answer : 4

88)

For spontaneous process

$\Delta G° < 0$, $K_{eq} > 1$, $E^0_{cell} > 0$

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89) **Explanation the Question:**
The question asks for a suitable indicator for the titration of formic acid (HCOOH), a weak acid, with potassium hydroxide (KOH), a strong base.
Concept:
This question is based on Titration of Weak Acid with Strong Base
Solution:
A suitable indicator must change color within this basic pH range.
• Phenolphthalein has a pH transition range of 8.2–10, which is basic.
• Methyl orange and Methyl red have transition ranges in the acidic region.
Therefore, phenolphthalein is the correct choice.

90) –12250x – 13000 (1 – x) = –12500
750x = 500 ⇒ x = 2/3 and y = 1/3
So, $\frac{x}{y} = \frac{2}{1}$

BIOLOGY

91) NCERT-XI, Pg. # 143, 144, 145

92)

NCERT-XI, Pg. # 141

93) NCERT-XI, Pg. # No. 145

94)

NCERT-XI, Pg. # 138

95)

NCERT-XI, Pg. 142

96) NCERT-XI Pg # 142

97)

NCERT-XI, Pg. # 139-141

98)

Module

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99)

NCERT-XI, Pg. # 140

100)

NCERT-XI, Pg. # 145

101)

NCERT-XI, Pg. # 159

102) NCERT-XI, Pg. # 159

103)

NCERT-XI, Pg. # 154, 155

104) NCERT-XI, Pg. # 155-156

105) NCERT-XI, Pg. # 157

106) NCERT-XI, Pg. # 163

107)

NCERT-XI, Pg. # 175

108) NCERT-XI, Pg. # 176

109) NCERT-XI, Pg. # 176, 177

110) NCERT-XI, Pg # 166
Life begins by seed germination and ends with senescence.

111) NCERT-XI, Pg. # 170

112) NCERT-XI, Pg.# 177

113) NCERT-XI, Pg # 66

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114) NCERT-XI, Pg # 60

115) NCERT-XI, Pg.# 63-64.

116) NCERT-XI, Pg. # 66

117)

NCERT-XI, Pg. # 64

118) NCERT-XI, Pg. # 65
Tomato, Lily and Lemon

119) NCERT-XI, Pg. # 61,63,64.

120) NCERT-XI, Pg. # 73,74,75,77

121) NCERT Pg. No. # 74

122) NCERT-XI, Pg. # 77

123) OLD NCERT-XI & Module

124) NCERT-XI, Pg. # 72, 73, 74
Cortex, Pericycle, Pith, Endodermis

125) NCERT-XI, Pg. # 76

126) NCERT-XI, Pg. # 72

127) NCERT-XII, Pg # 6, 7

128) NCERT-XII, Pg. # 22-23

129) NCERT-XII, Pg. # 5, 6

130) NCERT-XII, Pg. # 5, 6

131) NCERT-XII, Pg. # 18

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132) NCERT-XII, Pg. # 5

133) NCERT-XII, Pg. # 9

134) NCERT-XII, Pg. # 14

135) NCERT-XII, Pg. # 20

136) NCERT Pg # 317,318,333

137) NCERT (XII) ; Page No. # 321 (E)
NCERT (XII) ; Page No. # 321 (H)

138) NCERT XI Pg.#236

139) NCERT XI Pg.#236

140)

NCERT Pg. # 240-241

141) NCERT Pg.#243

142) NCERT Pg. # 243

143) NCERT Pg.#226

144) NCERT Pg. # 25,26,27

145) NCERT (XII) ; Page No. # 225

146) NCERT, Pg. # 46

147) NCERT, Pg # 36

148) NCERT, Pg. # 53

149) NCERT Pg. # 31, 33

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150) NCERT Pg. # 47

151) NCERT Pg. No. # 48

152) NCERT Pg.#231

153) NCERT, Pg. # 230

154) NCERT XI Pg.#245

155) NCERT XI Pg.#248

156) NCERT XI Pg # 217-218

157) NCERT (XII) ; Page No. # 30, 33

158) NCERT Pg # 34, 35

159) NCERT P.g No. 48

160)

NCERT Pg. # 231

161)

NCERT Pg. # 236

162)

NCERT Pg. # 232

163)

NCERT Pg. # 239

164)

NCERT Pg. # 242

165)

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NCERT Pg. # 243

166)

NCERT-XI, Pg. # 243

167) NCERT, Pg. # 245 to 247

168)

NCERT Pg. # 221

169)

NCERT Pg. # 217

170) NCERT Pg. No. # 220, 221

171)

NCERT Pg. # 227

172)

NCERT Pg. # 227

173)

NCERT Pg. # 227

174) NCERT Pg. # 44

175)

NCERT Pg. # 48

176) NCERT Pg. # 31

177)

NCERT Pg. # 34

178)

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NCERT Pg. # 34

179)

NCERT Pg. # 37

180)

NCERT Pg. # 30

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